DESIGN THINKING

THE GUIDEBOOK

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FOREWORD

The launch of this Design Thinking Guidebook for Public Sector innovation in Bhutan marks an important milestone in our public-service transformation journey. This guidebook is the culmination of many months of learning, applying and adapting the human-centred Design Thinking methodology and repertoires of tools.

The Royal Civil Service Commission in collaboration with Singapore Polytechnic, with funding support from Temasek Foundation International, Singapore and the Royal Government of Bhutan, organised the "Design Thinking for Public Sector Innovation Programme".

During the programme period from June 2016 till June 2017, the Design Thinking training was introduced to foster citizen-centric mind-set, and to embrace creativity and change in the public sector. Acquiring Design Thinking competencies and new ways of thinking can help generate innovative solutions to improve efficiency, quality, productivity, and user satisfaction. About 230 public service officers and leaders have undergone the Design Thinking training, out of which 23 have been further trained as Master Trainers.

The Design Thinking Guidebook was developed by the Design Thinking Master Trainers of Bhutan and facilitated by experts from Singapore Polytechnic. This guide will certainly go a long way towards creating an interest in Design Thinking as well as providing step-by-step instructions on how to apply the processes and tools in the context of public services. The guide book has certainly got the fundamentals right: it is simply written and illustrated with appropriate pictures to serve as useful references for both design thinking practitioners and beginners alike. The methods and tools have also been adapted for application in Bhutan.

The future is about public-sector innovation to meet new challenges. With the help of this guide, Bhutan public sector can learn to apply Design Thinking to nudge behavioral change, simplify and solve complex problems such as promoting Management Innovations like doing more with less, uncover strategic use of Human Resources and explore fresh backend syndication possibilities.

I am confident this Design Thinking guidebook will serve as an easy-to-use and effective guide to facilitate anyone to learn and apply the methods and tools to generate innovative ideas, and allow us to grasp the low hanging fruits in the short term and design better system in the long run.

Dasho Karma Tshiteem
Chairperson
Royal Civil Service Commission

BRIEF PROFILE OF SPI DESIGN THINKING PROGRAMME FACILITATORS

LEAD FACILITATOR

Mr Lee Chong Hwa has over 30 years career experience. He is currently a Design Thinking and education consultant with Singapore Polytechnic International (SPI). As a certified Design Thinking practitioner by Rotman DesignWorks, University of Toronto, he is a pioneer in infusing Design Thinking in Business Education. Chong Hwa was the architect of the SP Business School Design Thinking Masterplan. In this capacity, he was responsible for providing sound academic and professional design-thinking leadership in curriculum design, pedagogy, and training.

As a Design Thinking consultant, Chong Hwa has consulted, led and facilitated many Design Thinking projects and workshops in Singapore and overseas. Overseas programmes include the Philippines State Universities and Colleges, Ministry of Manpower, Indonesia and the Royal Civil Service Commission of Bhutan. Since 2014, Chong Hwa has been the principal trainer and facilitator for the Public Service Division (PSD) Design Practitioner Programme (a collaboration between PSD and Singapore Polytechnic).

An effective speaker, trainer and facilitator, Chong Hwa is skilled in employing clarity, captivating visuals, and motivation to deliver effective presentations and learning activities to diverse audience at all organisational levels. As a consultant, he has proven track record in developing innovative solutions through creative approaches, team orientation and effective execution.

FACILITATOR

Mr Jason Leow does his best work at the intersection of design and public good. While he has extensive experience in public innovation design consultancy for governments and social impact non-profits, he also aspires to create new products and services that deliver public value.

As a designer-turned-civil-servant leading cross-agency projects in the government previously, Jason offers an insider's perspective and a grasp of the hard trade-offs in public policy projects. He is deeply passionate to create social impact, blending entrepreneurial resourcefulness, empathy and creative confidence to make that happen.

FACILITATOR

Ms Jun Lau came with a strong background in organization development and leading service innovation projects. Her experience spans from working with client organizations from various industries including financial services, engineering, hospitality, healthcare, life sciences, manufacturing, security services, telecommunications and government agencies.

Jun leads projects to enhance organizational effectiveness through the use of change management principles and design thinking methodologies. In recent years, she has worked closely with leaders across levels, to solve design challenges through human-centred research and experience design. As a service industry judge and facilitator, Jun inspires teams and individuals to collaborate and co-create solutions for a better tomorrow. Jun believes that good design can change the world and impact lives for good. She enjoys facilitating leadership conversations in an intimate yet impactful way and travels to major cities in Asia for her regional projects.

FACILITATOR

Ms Angela Ho is an experienced coach, trainer/facilitator and Design Thinking practitioner. Her dream is to be a positive force for greater good. She wants to help organizations be more innovative, user-centric and agile, in the volatile, uncertain, complex and ambiguous (VUCA) business landscape and operating environment today.

Angela creates value through creating, executing and sustaining changes for organizations. Angela's approach is founded on a people-centred leadership model. She coaches teams to become high-performers and individuals to reinvent themselves, using a unique blend of coactive and agility coaching, design thinking, systems thinking, enneagram and mindfulness approaches.

FACILITATOR

Mr Chia Hui Yong has worked extensively in both public and private sectors in Singapore and the region, including social services, NGO, service and retail, education and training programme design and management. He has successfully led and executed many international humanitarian aid and international development projects.

Hui Yong is naturally curious and enjoys working with people across cultures. He believes Design Thinking has the potential to unlock creativity to innovate solutions for wicked problems and that embracing empathy is key to connect with people to forge common grounds and co-create solutions.

INTRODUCTION

Design Thinking?

Design Thinking believes (and has been proven) that the people who face problems are the ones who hold the key to their problem's answer.

Design Thinking is Human-centered problem solving tool which emphasize on Empathy, Collaboration, Co-creation and Stakeholder feedback to unlock Creativity and Innovation, which devises feasible and viable Big Idea/solutions. The key to the process is empathizing with the users (citizen) to uncover unmet needs by understanding their beliefs, values, motivations, behaviors, pains, gains and challenges and to provide innovative solution concepts.

Purpose of this guidebook

Equip the DT practitioners to work on real projects by designing needs and solutions with communities (users/citizen), to deeply understand the people they're looking to serve, to dream up scores of ideas, and to create innovative new solutions rooted in people's actual needs using Design Thinking through step by step guide with samples and ready to use templates. This guidebook offers a comprehensive set of Methods & Tools and activities that will take you from framing up your design challenge to getting it to the decision makers for implementation and users.

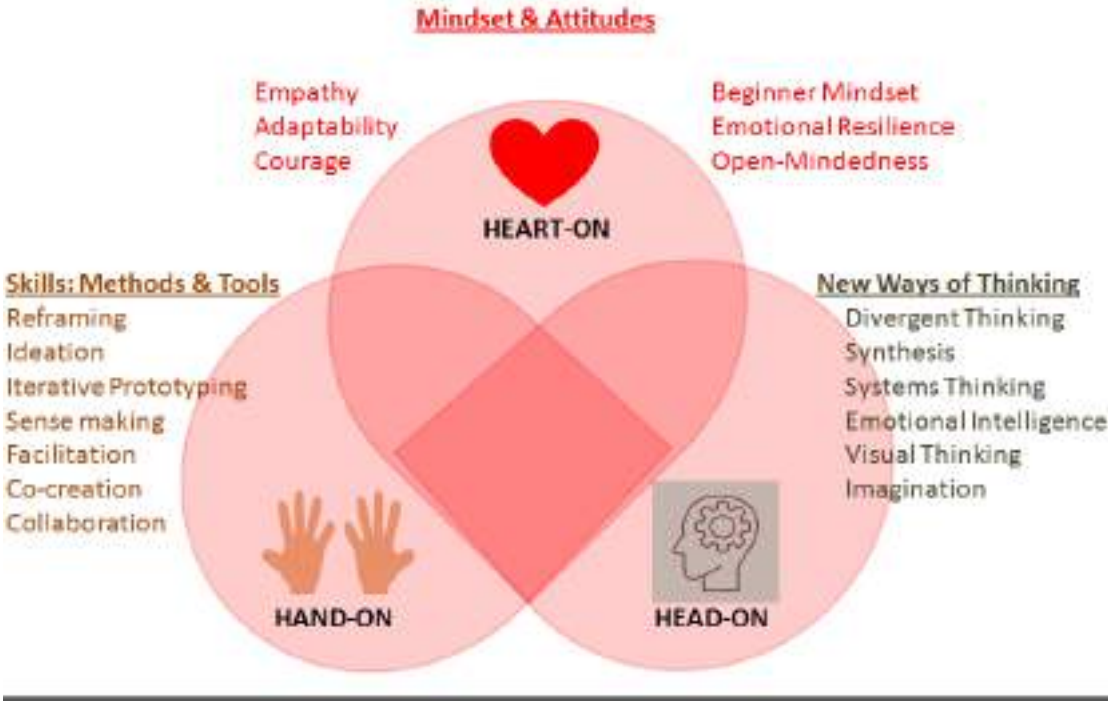
This guide book is a learning tool for DT beginners.

KEY PRINCIPLES AND MINDSET

Design Thinking human-centered problem solving approach is based on a few easy-to-understand principles:

Mindsets, Skills and Thinking

Heart-on Hands-on Head-On Approach



5 PHASES, METHODS & TOOLS OF DT

Module	Purpose	Phases	Methods and Tools	Pages
INSPIRATION (Problem Framing)	Reframe Opportunity	EXPLORE	STEEP Analysis	pg 5
			Strategic Priorities	pg 7
			Activity System	pg 9
			Stakeholder Mapping	pg 11
			Opportunity Framing	pg 13
	Rediscover Your User Deep Needs	EMPATHISE	Observations	pg 19
			Need Finding	pg 25
			User Personas	pg 30
IDEATION (Problem Solving)	Refresh Your Solutions & Innovative Ideas	EXPERIMENT	Ideation	pg 37
			Prototyping	pg 47
	Reaffirm Your User Needs	ENGAGE	Storyboarding	pg 54
			Co-Creation	pg 55
			Idea Refinement	pg 23
IMPLEMENTATION (Solution Testing)	Review Your Activities & Strategies	EVOLVE	Concept Synthesis	pg 62
			Strategic Requirements	pg 64
			Evolved Activity Systems	pg 66
			Quick Wins	pg 73

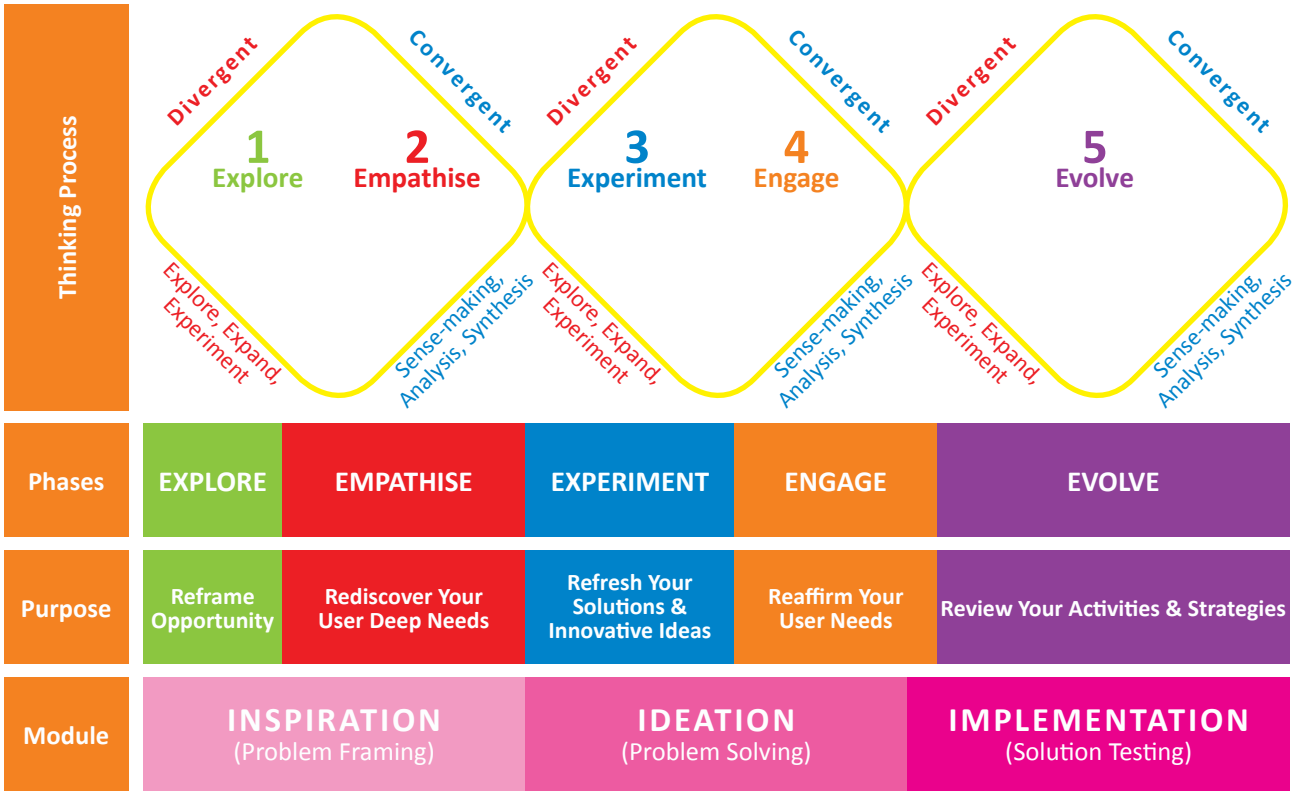
USER GUIDE

How to Learn and Apply the Design Thinking Process, Methods and Tools

Design Thinking is not a perfectly linear process, and each project invariably has its own contours and character. Generally, you will move through three main modules: Inspiration, Ideation, and Implementation which includes 5 phases. When you move through the 5 phases, you will move from divergent thinking (create choices) to convergent thinking (make choices).

By going through these phases in turn, you will build deep empathy with the communities (user/citizen) and individuals you are designing for, figure out how to turn what you have learned into a chance to design a new solution and co-create and test your ideas before finally putting them out into the world. You can also follow specific phase or a Method & Tool individually at your workplace to solve a problem.

Though no two Design Thinking projects are alike, you can use the same Methods and Tools for each of the phases for every project. For example, to maintain creativity and energy, you work in teams, to build deep empathy with the people you conduct interviews, to keep your ideas visual, practical and relevant, you make tangible prototypes and since you rarely get it right the first time, you share what you have made and iterate based on the feedback you get through co-creation.



FOUNDATION BUILDING FOR DESIGN THINKING PROJECT

Scoping of design challenge

When you want to work on real project using Design Thinking, begin the process with a specific and purposeful situation or problem to solve called Design Challenge. The Design Challenge should be doable, clear, actionable and hopefully inspiring. Your problem statement should be an outcome and not a solution.

You can use SCOPES as a tool to brainstorm and frame your problem.

SCOPES		DEFINITIONS
S	SITUATION AND/OR PROBLEM	What is the **SITUATION** or **PROBLEM** that you are trying to address? • .....
C	CONSTRAINTS	What are the **CONSTRAINTS** when trying to address the situation or problem? • .....
O	OBJECTIVES AND OUT-COMES	What are the **OBJECTIVES** and intended **OUTCOME** of this design challenge? • .....
P	PEOPLE	Who are the **PEOPLE** that you are trying to help? Whose problem are you trying to address? • .....
E	ESTIMATES	What are the **ESTIMATED** resources required for this design challenge? • .....
S	SCOPE	Is there ample **SCOPE** to explore multiple alternatives to address the problem? • .....
DESIGN CHALLENGE		Frame the problem statement into "**How Might We...?**" • .....

PHASE I

explore

PHASE I EXPLORE

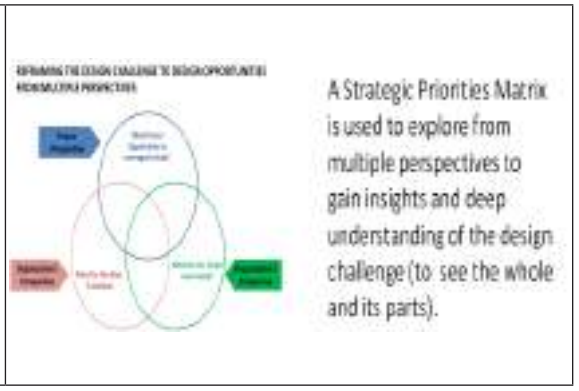
After scoping the design challenge, the explore phase focuses on reframing the design challenge topic into design opportunity from multiple perspectives.

The explore phase will help you:

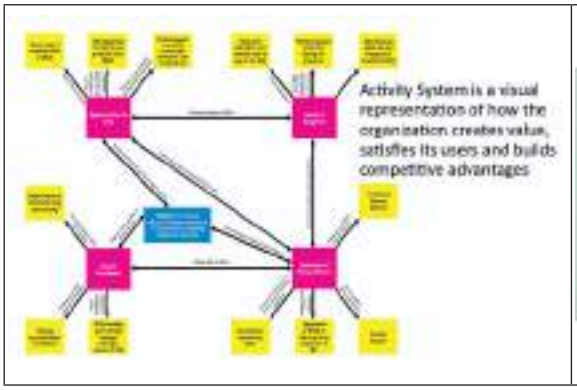
- Synthesis the STEEP trends analysis to gain insights on the implications and context of your design challenge.
- Foster multiple perspectives to explore your design challenge.
- Map the organization's activity system/ecosystem as the foundation model to leverage for your new idea delivery.
- Map key stakeholders to appreciate the key people who determine the success of your design challenge
- Frame project (design challenge) into design opportunity
- Identify, select and invite your target stakeholders for the interview
- Plan your design challenge project management.

METHODS & TOOLS

STEEP Analysis



Strategic Priorities



Activity System

Stakeholder Mapping

Opportunity Framing

STEEP ANALYSIS

Once you know your design challenge, you need to understand the context of your design challenge which can be done through STEEP analysis to determine the impacts of macro environmental trend related to the design challenge to understand what possible (possible innovative ideas).

What is STEEP Analysis?

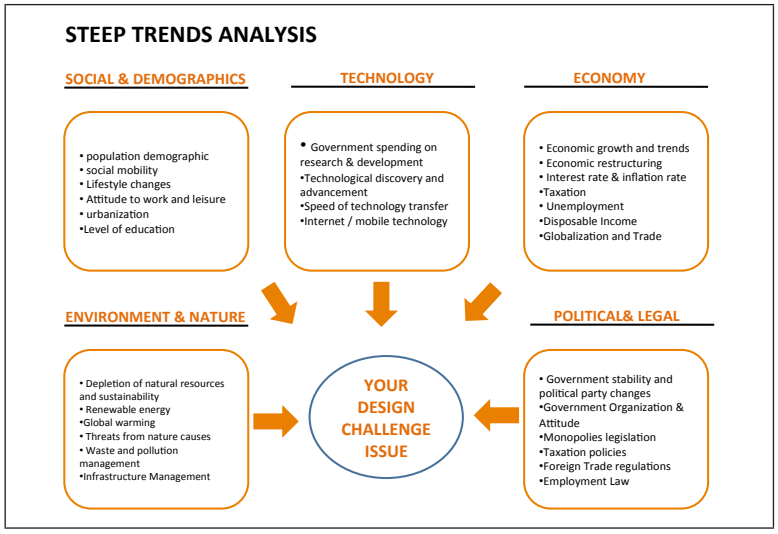
STEEP Analysis is a tool to explore and determine the impact of macro-environmental trends in the context of the project topic (design challenge) as you should not limit your thinking just to the people you are designing for. You may need to consider governments, international partner and other stakeholders. STEEP Analysis is used:

- To understand the future opportunities and challenges.
- To keep an eye on the future while focusing on the possibilities of the current as new services, processes, administration and public policies may have to be developed in response to these trends.
- To cultivate thinking which leads to future implications of the present changes?

How to conduct STEEP Analysis

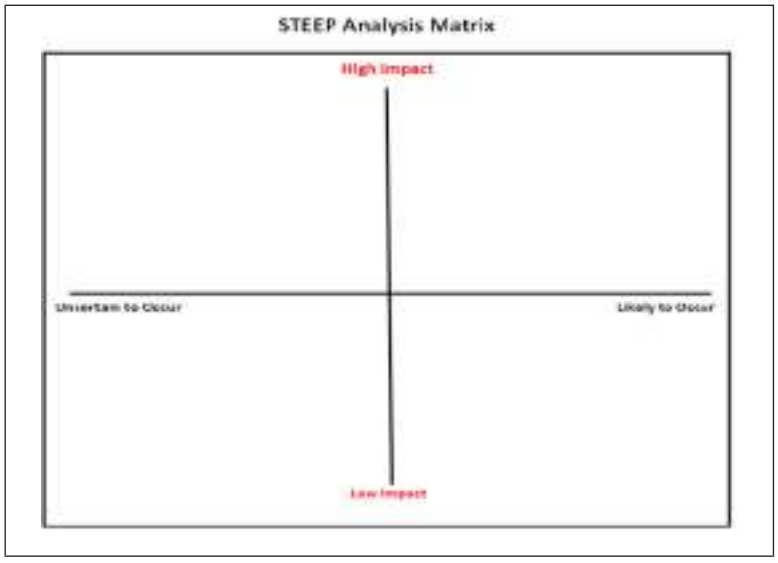
Step 1	• Reframe the design challenge title and understand the design challenge statement in depth
Step 2	• identify the relevant trends affecting the design challenge by studying secondary data to back up the analysis
Step 3	• Compile the trends based on impact and (un)certainty of its occurrence for each trend and evaluate them as **Opportunities** or **Challenges**
Step 4	• Review the entire analysis to ensure that trends and implications identified are relevant to the design challenge

Templates to be used for the conduct of STEEP analysis



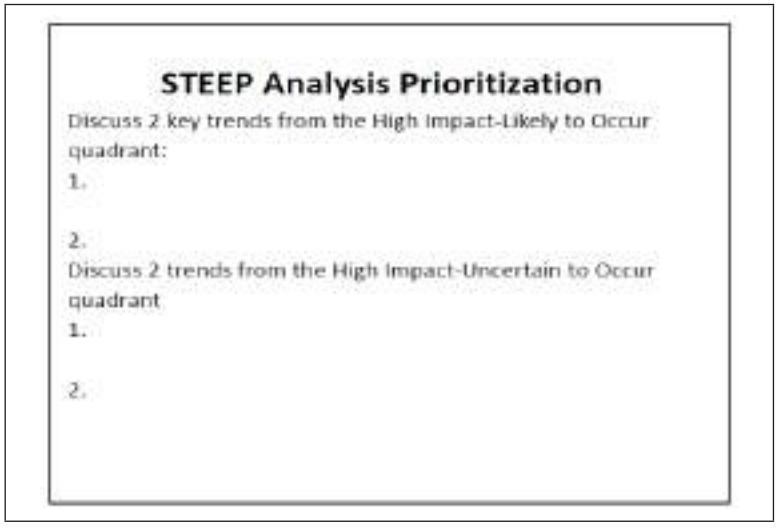
STEEP TRENDS ANALYSIS TEMPLATE

List down trends under each category and label them as opportunities and challenges



STEEP ANALYSIS MATRIX

Sort the trends based on its impact and (un)certainty of its occurrence.



STEEP ANALYSIS MATRIX PRIORITIZATION TEMPLATE

After mapping the trends in the matrix, filter few trends using this template.

STRATEGIC PRIORITIES

This tool is used after the STEEP Analysis. A Strategic Priorities Matrix is used to explore from multiple perspectives to gain insights and deep understanding of the design challenge (to see the whole and its parts).

What is Strategic Priorities?

Strategic Priorities Matrix is a tool with which you will look through multiple perspectives to better understand the context and the real issues with your design challenge and help you synthesise and formulate a point of view or theory to explain your design challenge problem.

How to use the Strategic Priorities

Step 1	• Revisit the filtered STEEP trends and look for patterns, links, relationships, cause and effects and correlation amoung the research findings leading to new insights and fresh perspectives of your design challenge
Step 2	• Understand the reality that focuses on the relationships amongst the parts of the system and the dynamics those relationships produce
Step 3	• List what challenges need immediate attention going forward for your design challenge and why?
Step 4	• Identify which trends are more important than others
Step 5	• Synthesize the trends and trategic priorities based on importance of the relationships amoung the trends, opportunities and challenges that demand immediate attention for key stakeholders

Templates for conducting Strategic Priorities.

STRATEGIC PRIORITIES MATRIX

	URGENT	LESS URGENT
IMPORTANT		
LESS IMPORTANT		

STRATEGIC PRIORITY MATRIX

From the STEEP Analysis Matrix prioritization Template, filter and the trends using this template

SYNTHESIS: MAKING SENSE OF STEEP ANALYSIS AND STRATEGIC PRIORITIES

Assessment Questions	Synthesis: Sense Making
What relationships among the trends do you perceive? How are they related? Why are these relationships important?	
What opportunities and/or challenges need immediate attention going forward for your design challenge? And why?	
What would it take to create positive change on this issue relating to your design challenge?	
Who else wouold be interested in this issue? Why should they care? What conversations would you have with them?	

SYNTHESIS: MAKING SENSE OF STEEP ANALYSIS & STRATEGIC PRIORITIES TEMPLATE

ACTIVITY SYSTEM

You can use the activity system to gain insights on the organization's strengths, weaknesses, advantages and gaps to explore fresh opportunities and possibilities. It should also be used to establish foundation for leveraging and evolving the strategic 'ecosystem' or a model in the Design Thinking Phase 5: Evolve.

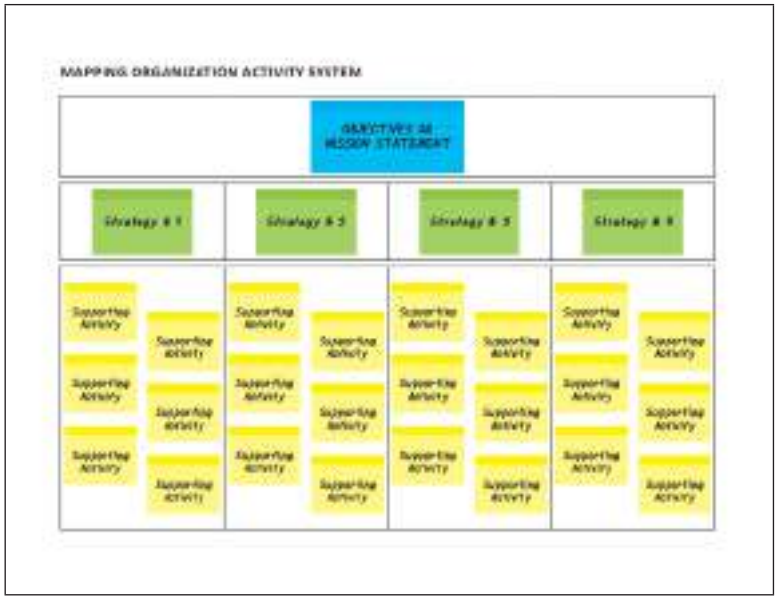
What is Activity System?

Activity system is a visual representation of how the organization creates value, satisfies its users and builds competitive advantages. It also identifies gaps in achieving the objectives (providing service) and possible duplication of jobs. A powerful organizational model is one with a unique activity system that has mutually reinforcing activities which is difficult to replicate.

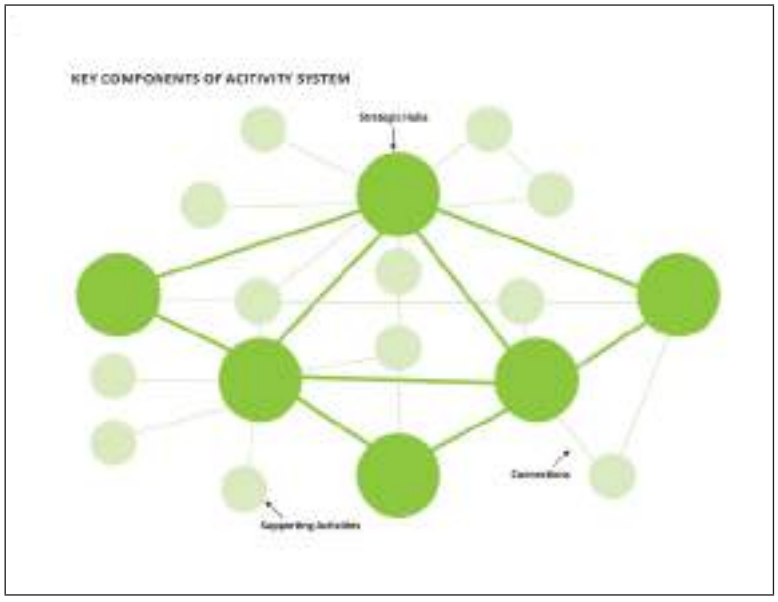
How to define the Activity System

Step 1	• Identify the vision, mission, goals and objectives of the organization
Step 2	• List all current activities carried out in relation to the dsign challenge and identify the key strategies (Strategic Hub)
Step 3	• List all the support activities for each strategy (Strategic Hub)
Step 4	• Determine the relationships between the strategic hubs
Step 5	• Determine the relationships between supporting activities and the strategic hubs
Step 6	• Organize/distill the inputs and map the activitiy system

Templates to map activities



ACTIVITY SYSTEM MAPPING TEMPLATE



KEY COMPONENTS OF ACTIVITY SYSTEM

STAKEHOLDER MAPPING

Stakeholder Map can be used to help you visually summarise and communicate the relationships between various stakeholders when working on a design challenge or any issue that require to understand various players involved.

What is Stakeholder Mapping?

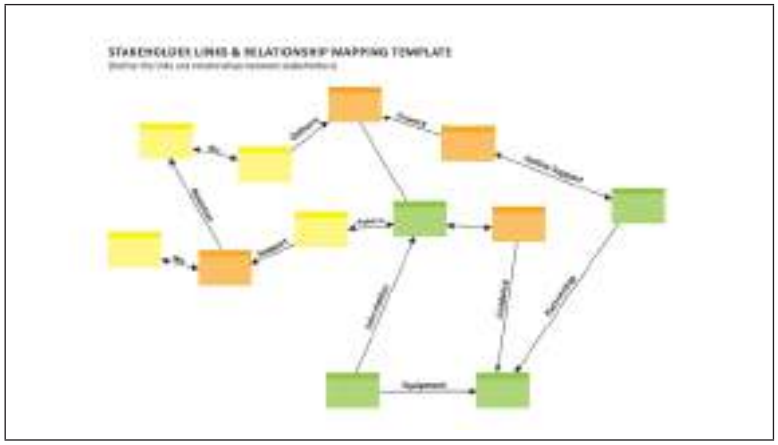
Stakeholder mapping is a process to identify the key people, group and organization that have direct influence on the design challenge and understand the key stakeholders so as to better engage them. It draws the relationship between the stakeholders and shows the importance of the stakeholder(s) as it plays a key role in influencing the development and outcome of the design challenge.

How to conduct the Stakeholder Mapping

Step 1	• identify relevant groups, organizations and people within and outside the organization. agency who has interest/influence on the design challenge
Step 2	• Plot the stakeholders on the matrix based degree of interest and influence through understanding their perspectives and unique interests on the design challenge
Step 3	• Cluster the stakeholders based on Central Agency, People, Local Government
Step 4	• Analyse and identify the link and relationship among these key stakeholders
Step 5	• Develop strategies to engage these key stakeholders: Design actionable strategies to garner the support of key stakeholder relevant to the issue (design challenge topic)

Templates to conduct Stakeholder Mapping

STAKEHOLDER MAPPING MATRIX



STAKEHOLDER LINKS & RELATIONSHIP MAPPING TEMPLATE

STAKEHOLDER PRIORITY MAPPING MATRIX

STAKEHOLDER ANALYSIS & ENGAGEMENT STRATEGY TEMPLATE

The last step is to synthesize the above matrices and template using this template to come up with the Stakeholder Engagement Strategy.

OPPORTUNITY FRAMING

Opportunity Framing is carried out after you know and understand the 'Future Trend', 'Organization' and the 'user' (stakeholder) to relook into the design challenge and reframe into opportunity.

What is Opportunity Framing?

Opportunity Framing prepares you to better define the design challenge, ensuring all relevant notes have been considered before reframing the design challenge and transform the design challenge into design opportunity with many possibilities (It is not only about thinking out of the box but it is about the expansion of the box). This will also help you in identifying the potential interviewees for later phases.

"How might we <verb> a <desired end state/outcome/issues for <user>?

How to do Opportunity Framing

Step 1	• Revisit the initial design challenge (issue/statement), the STEEP analysis, strategic priorities, activity system and stakeholder mapping
Step 2	• Look at the design challenge from various perceptive i.e. the Future Lens (STEEP analysis and strategic priorities), Organizational Lens (current activity system) and User Lens (stakeholder mapping)
Step 3	• Use project brief and opportunity framing assessment questions as a guide
Step 4	• Discuss and answer the questions in the project brief scoping template. (1) challenge definition, (2) opportunity framing
Step 5	• Come up with 3 to 4 alternative "How might we...?" from opportunity/possibility statements and narrow it down to one

The fundamentals of Opportunity Framing are:

1. From the industry focus to user focus (Public Sector Agency to Citizen or End Users)
2. From single idea to system
3. From symptoms to root cause (What are the real issues, pain points, underlying deep needs)

Templates to be used for the Opportunity Framing

PROJECT BRIEF AND OPPORTUNITY FRAMING TEMPLATES

PROJECT BRIEF AND OPPORTUNITY FRAMING TEMPLATES

PROJECT BRIEF AND REFRAMING PROJECT CHALLENGES TEMPLATES

PHASE II

empathise

PHASE II EMPATHISE

Empathise is a mechanism to understand and share the feelings of your users to foster deep user understanding and be able to uncover the deep user insights and needs.

Purpose:

- to foster empathy and deep understanding of the users in terms of their life, needs, aspirations and challenges;
- appreciate others as human beings and understand and relate with their feelings;
- see their world through their eyes and make emotional connection;
- communicate understanding with others and share their world;
- be in the shoes of your users, experience same and gain fresh insights and uncover their needs; and
- Develop passion to act and help and inspire us to find new solutions.

Empathy is used to uncover the deep user insights and needs by gaining broader perspective of the citizens' life.

Methods & Tools, Mindsets & Process:

METHODS & TOOLS

FIELD OBSERVATION

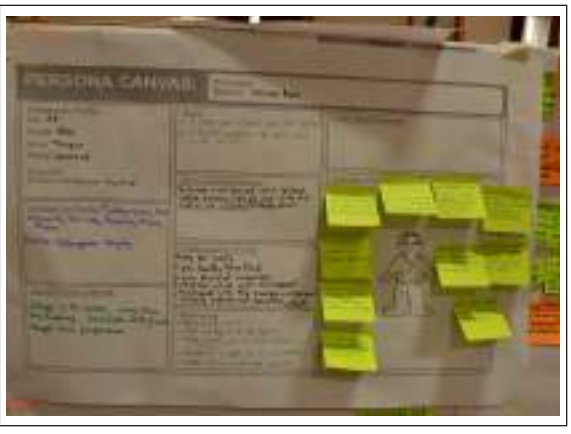
Structured approach to observing people in their natural Environment to uncover user insights and fresh perspectives of people and their behaviours.

DEEP USER INTERVIEW

An art of conversation to elicit stories and uncover deep user insights and needs – both latent and unmet needs.

NEEDS FINDING

Human process of making sense & transforming your observations and deep user interviews into usable data cluster &meaningful insights to uncover the unmet needs of your users.



PERSONA DEVELOPMENT

A process of humanising your target users, giving voice and character and making them real.

FIELD OBSERVATION

What is Field Observation?

It is a structured approach to observing people in their natural Environment rather than in a formal research setting. It is to uncover user insights and fresh perspectives of people and behaviour.

Methods to conduct observation:

Shadowing (A Day in the Life)	Fly on the Wall	Site visits	Body storming

When to use Field Observation?

Field observation is used to uncover User Insights and fresh perspectives of People and Behaviours.

How to conduct Field Observations

Approach your onsite observation with an open mind and observations need to be made with all your senses which include both first sight – observing observable as reality, and second sight (observing the unobservable). When conducting observation, one needs to be:

- Attentive (in the presence)
- Curious (keep asking why?)
- Perceiving with all your senses
- Open-minded to learning

Use POEMS framework to structure and guide the observation as presented in the POEMS Framework Template (Annex) to record and capture what you have observed.

To make sense of the observation, we need to look beyond the obvious and explore WHY than just WHAT in terms of relationships, behavioral patterns, interactions, gaps, and mistake. Take photos to support and document your learning as follows in relation to POEMS framework.

P		• **People:** Who are involved? (e.g. commuters, street vendors, office workers, children, motorists, delivery men, etc.) What roles do they play? How are the people engaging with each other? How are they related? What is the relationship? What is the social context?
O		• **Objects:** What artifacts are important? (e.g. bus stops, signage, traffic lights, benches, etc.) What roles do they play? How are people engaging with the objects, and with their surrounding? What is interaction? How does the object influence people's behavior?
E		• **Environment:** Where is the action taking place? (e.g. public bus, road side, government office, garden, etc.) What is happening? What are the people doing? How do the people behave in this environment? How does the environment influence people's behavior? What is the mood? Ambience?
M		• **Messages & Media:** What are the messages and communication media used? (e.g. signage, online materials, posters, apps, etc.) What roles do they play?
S		• **Services:** What are the services and support systems provided? (e.g. registration, library services, hands-on guide, online booking, etc.)

- **Context**: where, when, who and why?
- **Insights**: what have you learned about your target users
- **Challenges**: what are the pain-points, obstacles, gaps
- **Opportunities**: What are some opportunities and new possibilities
- **Perspectives**: How would you rethink about your design challenge project

Field Observation Preparation

Before embarking on your field observation, ensure the team has a well thought out plan as outlined below:

1. Who & Where to Observe? (Think about the profile of your target group(s)	**2. What are different roles of your team members?** (Interviewer, Observer, Note Takers)
3. What to Observe and learn? (Think about the issues, concerns your target group may have)	**4. Inventory check before setting off** (Observation list, report, template, camera, pen etc.)

DEEP USER INTERVIEW

What is Deep User Interview?

Deep user interview is an art of conversation to elicit stories and to uncover deep users' insights and needs – both latent and unmet needs through understanding of the users';

- Behaviour & feelings,
- Goals,
- Motivations,
- Aspirations,
- Values,
- Beliefs,
- Pains and challenges.

When to use Deep User Interviews?

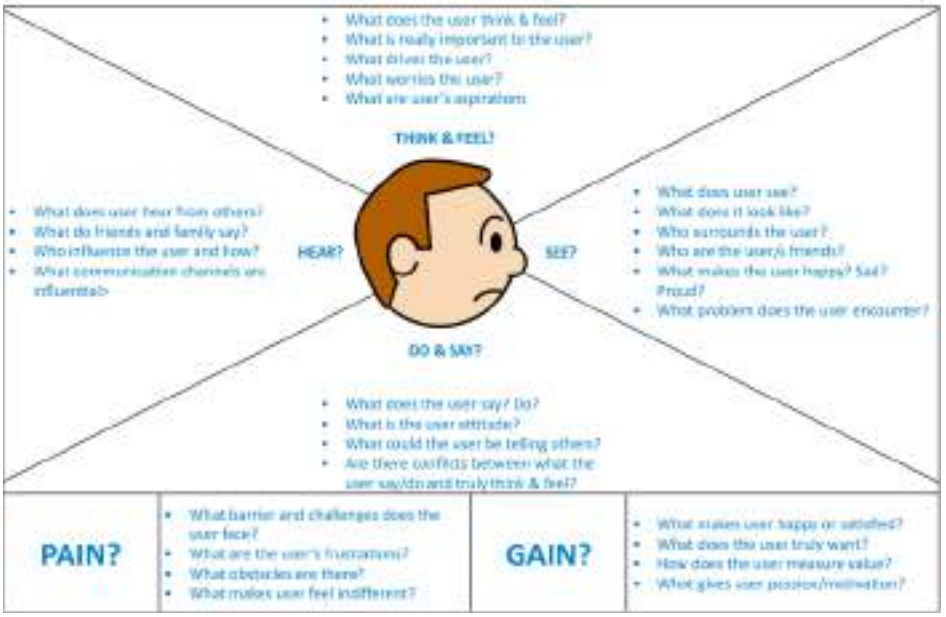
A Deep User Interview is conducted when we want to understand beliefs, values, challenges, concerns, frustrations, motivation & behaviours, and uncover user insights, latent needs, and the unmet needs of the users by listening to their stories and experiences. A proper conduct of deep user interview is also important for theme clustering, insight development, need findings, and persona development. Need findings and persona development depend on the quality of interviews that we conduct.

How to conduct Deep User Interviews?

1. Pre-interview preparation

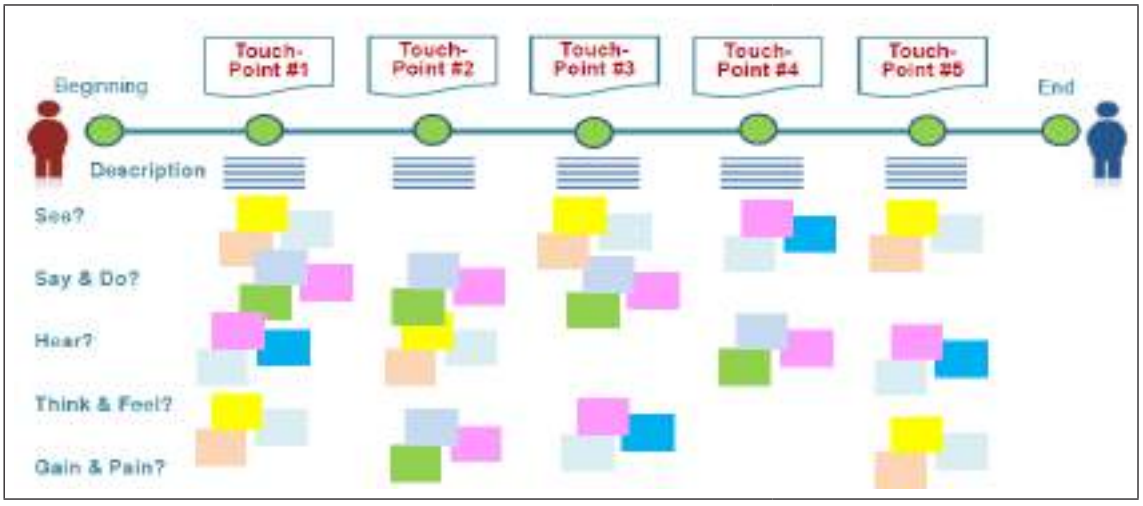
Being prepared with right questions and other interview tools and techniques are key to successful conduct of a deep user interview. Prepare your deep user interview by carrying out following activities:

- Prepare interview questions
  - Be prepared with the rights questions to explore the deep user needs with use of both Type A – closed ended and Type B – open-ended probing questions.
  - Use **EMPATHY Map** and/or **JOURNEY Map** to generate right questions as shown in Annex (***Empathy Map & User Journey to Generate Interview Questions***).
  - Use of empathy map helped to generate questions related to *think & feel, see*, *say & do*, *hear, pain* and *gain*.



Sample Empathy Map

Sample Journey Map



Combined Empathy & Journey Map to generate Interview Questions

- Plan and structure your interview
    - Plan and schedule your interview on a mutually convenient date and time.
    - Plan you interview for at least 90-120 minutes for each interview sessions. It takes time for the users to open up.

Structure your interview to evolve overtime with the use Type A questions followed by Type B and concluding with Type A.

Picture/word cards are given to interview

- Identify interview team and assign roles
    - An interview team should have an interviewer and a note-taker/observer.

Prepare your equipment check-list and relevant tools

- Simple interview tools such as *photos* and *picture cards*, *words cards, journey map*, and *5 Whys* will help interviewee to open up.

- Conduct mock interview
    - The objective is to familiarize team members with the interview process.
    - Conducting deep user interview takes a lot of practice.

2. During the Interview

Conducting a proper interview with the user is key to getting the rich and deep insights. User insights are sources to understanding the user unmet needs.

- Follow your interview structure – use of Type A and Type B questions - and make it a natural and a causal chat.
- To begin the interview, use Type A questions relating to the demographics and habits to build rapport and make interviewee comfortable.
- To explore, elicit stories and deeper response and gather information on personal motivation such as aspirations, inspirations, motivations and pain points use Type B questions.
- Use Type A questions to gather information related to the project statement. These questions were prepared to be asked towards the end of the interview to wrap up. During
- Avoid questions that lead to a dead end. Use interview tools and techniques to probe more and evoke stories and explore emotions.
- Be comfortable with silence and observe for non-verbal cues and emotions.
- Use ***User Interview Notes Template*** (refer annex) to record everything in verbatim. Do not interpret or analyse anything during the interview.
- Take photo references of the interview process, including activities like sketching, journey mapping, card sorting, etc.

There are five main activities of conducting Deep User Interview:

1. **Ask** - asking right questions (ask open-ended probing questions),
2. **Listen** – listen for deeper meaning, listening with purpose (empathic listening – listening with all **senses** – and wonder why that is important),
3. **Observe** – observing with all senses,
4. **Sense** – make inference to gain clarity, and
5. **Record** – record everything in verbatim.

FIVE MAIN ACTIVITIES OF DEEP USER INTERVIEWS

ASK	LISTEN	OBSERVE	SENSE	RECORD
Ask Open Ended	Listen for Deeper	Observe with All	Make Inference	Record Everything

3. **Post Interview**

- Conduct post-interview debrief immediately after each interview sessions. Use Post ***Interview Discussion Template*** (annex) to summarize what you heard during the interview and develop a common understanding about the user interviewee.
- Conduct post-interview debriefs presentation to the team. Use post-interview ***De-brief Presentation Templates***.

Sample Post-interview De-brief Presentation

NEEDS FINDING

What is Needs Finding?

Needs finding is a deep human process of making sense and transforming your observations and Deep user interviews into usable data cluster and meaningful insights to uncover the unmet needs of your users (citizens). When conducting needs finding you are looking for patterns and relationships in terms of SPICE and SAM (Think-Feel-Do) frameworks to uncover Insights, Aspirations, Motivations, Challenges, Pain Points, and Deep Needs.

When to use Needs Finding?

The needs finding process is used to convert observations and deep user interviews into usable data cluster from which meaningful insights and needs of the users are discovered.

How to Find Needs

Step 1	• Have the interviewer read out the interview notes to the team as a "first person" while others capture key quotes, high and low points, and observations. The members must capture one data per post-it
Step 2	• Take turns to share what you have captured on your post-its and cluster common patterns and/or consistent issues emerging or compelling insights
Step 3	• Have a in-depth discussion for each cluster and examine to uncover deep USER INSIGHTS/NEEDS. This is a deeply human process that requires you to rely on your sensing and keep asking WHY until you uncover the real motivation.
Step 3	• Needs finding process is like a 'Map Making'. You refine your 'user-map' as you dive deeper and gain better understanding of your USERS. Remember that there is no right Map and is not about finding the correct answer. It is about Mapping an emerging picture of your Users as the team collaborate, compare and gather diverse views to make sense of interview findings and clusters or themes. To uncover the deep user needs, you have to first uncover the USER INSIGHTS.

Uncovering USER INSIGHTS

User insights are not user needs but it helps to explain it. Good user insights reveal a deep discovery about your user that can be acted upon to creating new ideas and solution. User insight finding takes times and some serious probing and trying to understand the real 'why' of the behavior.

5 Principles of uncovering User Insights:

1. User insight is a human truth
   A human truth that is moving and relatable. It connects to who we are as humans and why we feel what we feel.
2. User insight digs deeper than observation
   Help uncover the hidden motivation behind a behavior
3. User insight offers new, untapped understanding
   Offer undiscovered truths of the human complexion, something new, true and not obvious.
4. User insights shapes new perspectives and challenges old assumptions
   Shape your perspectives and challenge you to question assumptions.
5. User insight serves as a foundation to unlock creative thinking
   Give you the "aha" moment that inspire creative innovation.

How to uncover User Insights:

Look Everywhere; Recognize Relationship; Seek the Wisdom of "WHY?"

Recognise Relationship:

Connections – Tensions; Contrasts.

Events – Success & Failure; Good & Bad.

Cause & Effect – A leads to B; Necessary but not sufficient.

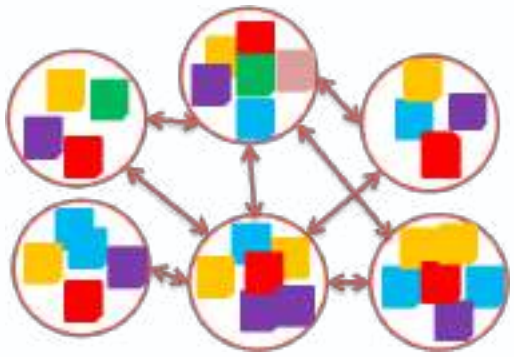
Gap – Mismatch; Misaligned; Lacks; Frustrations; Pain Points.

Timeline – Ahead of plan or behind time or deadline.

Outcome – Planned or Unintended.

Speed – A process sped up or slowed down

Search Everywhere: look beyond obvious; stories, events, events, experiences, behavior, goals, motivations, challenges, frustrations, tensions, pain-points, etc.



Seek the Wisdom of Whys: Ask thoughtful questions:
Team to have open conversations, put oneself in interviewee's situations, keep asking "why", map the thoughts, capture the insights on post-its.

Look for emerging patterns: Recurring Points or themes; similar situations; common experiences; cycle of events (vicious or virtuous); intensity of the experience

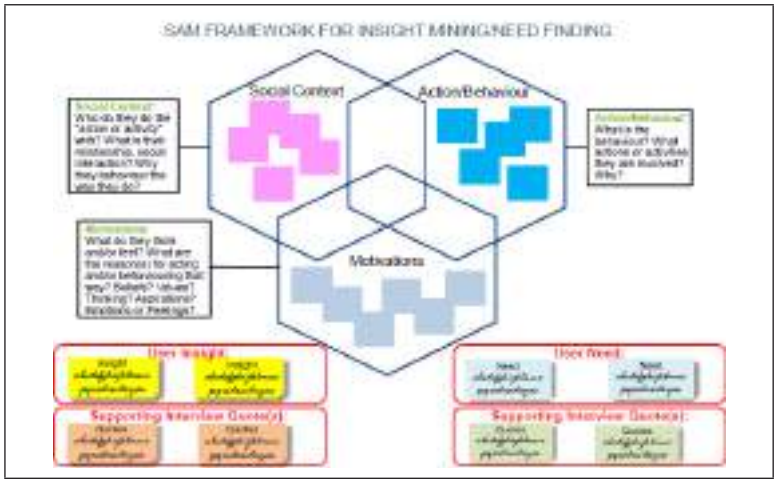
Understand the Action and Behavior:
Perceive the Social context, Sense the Motivation, Feel the Emotions.
Recognise the types of relationship

Tool to Uncover User Insights: SAM (Think – Feel – Do) framework

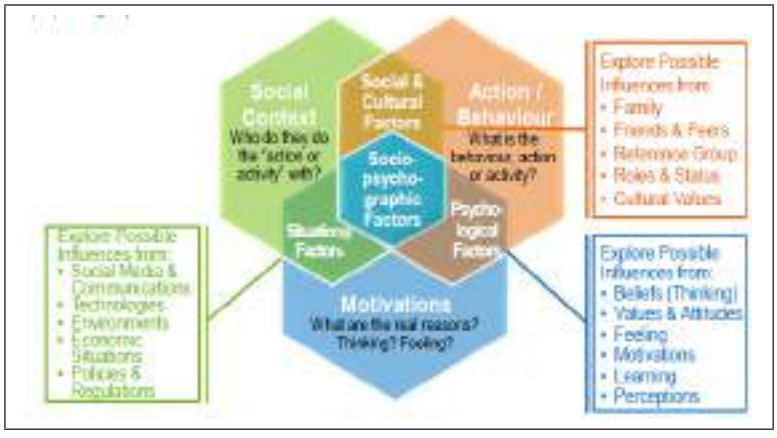
Social Context: who they do the 'action or activity' with? What is their relationship, social interaction? Why they behave the way they do?

Action (Bhaviour): What is the behavior? What actions or activities they are involved? Why?

Motivation: what do they think? How do they feel? What are the reason(s) for actions and/or behaving that way? Belief? Values? Thinking? Aspirations? Emotions or Feelings?



SAM Framework for Insight Mining/Need Finding



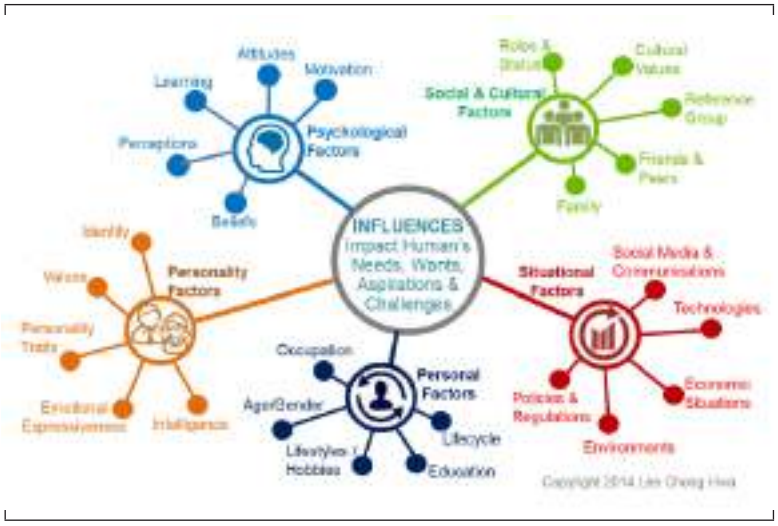
Uncovering User INSIGHTS to User NEEDS (SAM Framework)

Uncovering User Insights to User Needs:

Need is something that is missing. Need finding process is about looking through the situations or condition to find what is that 'something' missing for the target users. Need is a verb not a Noun. Needs are explicit or implicit (latent or unmet) needs. User Needs finding is a process or uncovering implicit, latent or unmet needs of the users. Latent or implicit needs are that the Users have but unaware or have not yet directly recognized, hence they are unable to clearly articulate. Latent needs are harder to find and defined but they are important source of inspiration for innovation.

Uncovering latent or unmet needs of the users require understanding the deep human needs and have to be drawn from the User Insights using S.A.M and SPICE Needs framework.

Understanding & Uncovering Deep User (Human) Needs: SPICE Needs Framework



Patterns, Relationships & Systems Ecology of Human Needs

SPICE Needs Framework

S	• **SOCIAL**: What does this person need form relationships with the people around them? (e.g. trust, qualified counsel, resourceful service staff, etc.)
P	• **PHYSICAL**: What does this person need on a functional and practical level? (e.g. responsive service, conducive environment, access to amenities, etc.)
I	• **IDENTITY**: What does this person need to define themselves? (e.g. to be contributor, raise civic consciousness, etc.)
C	• **COMMUNICATION**: What information does this person need? (e.g. non-bureaucratic language, access to reliable information, responsive feedback channels, etc.)
E	• **EMOTIONAL**: What does this person need emotionally and psychologically? (e.g. sense of national pride, sense of fairness and justice, etc.)

Sample Needs Statements

I need to regain my lost time at waiting to spend them meaningfully.

I need to overcome my self-doubt so that I can excel in my studies.

I want to be role model for my children and others.

I need a sense of familiarity so as to explore the unfamiliar.

People want to have that sense of belonging to a community to know that they belong.

During weekend I want to recapture quality time with my loved ones.

To be recognised for my efforts in the organisation.

Examples of Needs Statements

Documenting User Insights and Needs

It is important to document deep User Insights and Needs. User needs statements to be framed as verb – i.e. activities or desires with which your user could use help, not nouns (solutions).

INSIGHT #3:

While students instructors relationships are positive, students want instructors to exhibit good discipline

Narrative:
Instructors prepare trainees to improve performance in its activities mainly on work attitudes in working on the workpiece in practice and skills through learning on the job-related behaviours, knowledge, and expertise of the participants. Standard time is a measure in the assesment and implementation of competency-based training, in accordance with the curriculum in the training program.

Need:
Students look up to instructors as role model.

Quote 1:"There's always instructors that are not available when students need to ask questions. Opportunity to ask questions about our tasks are always delayed and delayed further."

Examples of Needs Statement documentation

PERSONA DEVELOPMENT

What is Persona Development?

Persona development is a process of humanizing your target user, giving voice and character and making them real. Personas are fictional characters created to represent your group of target users who exhibit similar behaviors, patterns, motivations and goals. A well created persona provide realistic and detail descriptions, that include Behaviors, Patterns, Motivations, Goals, Skills, Attitude, Challenges, and environment with a few fictional personal details, including a name, to bring personas to life.

When to use Persona Development

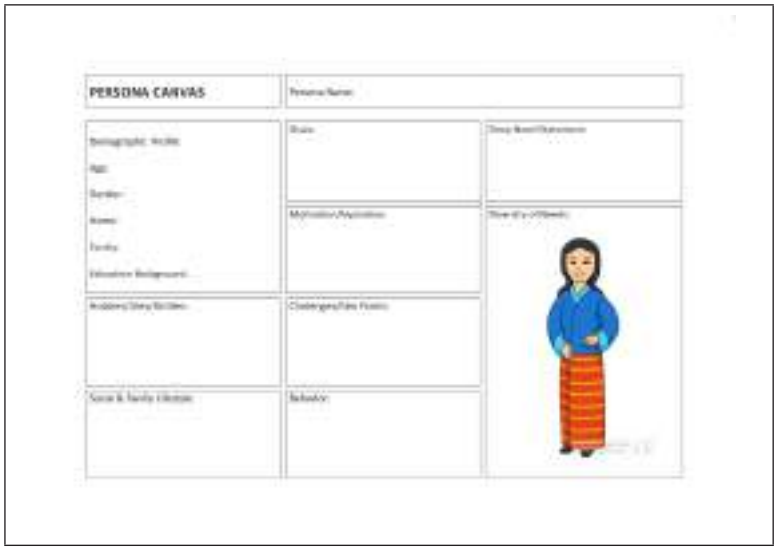
Persona development is to create user models – characters with a clearly defined purpose and characteristics – who will represent your target users throughout the design process from brainstorming for solution ideas to designing ideal user experience journey. Development of personas support story-telling, foster user understanding and evolve design. Personas tell stories and stories are part of every community. Stories help organize and convey information in a compelling manner and evoke emotions and responses.

How to Develop Persona

User personas are distilled from your observations and deep user interviews. Personas are developed as follows:

Step 1	• Review all your clusters, user insights and user needs from across your user interviews.
Step 2	• Distill those information relating to behavior patterns, goals, motivations, challenges, pain points, needs etc.
Step 3	• Add fictional personal details such as name to bring the persona life.
Step 4	• Add some narratives or story to reveal the persona's lifestyle, activities, choices and social context.

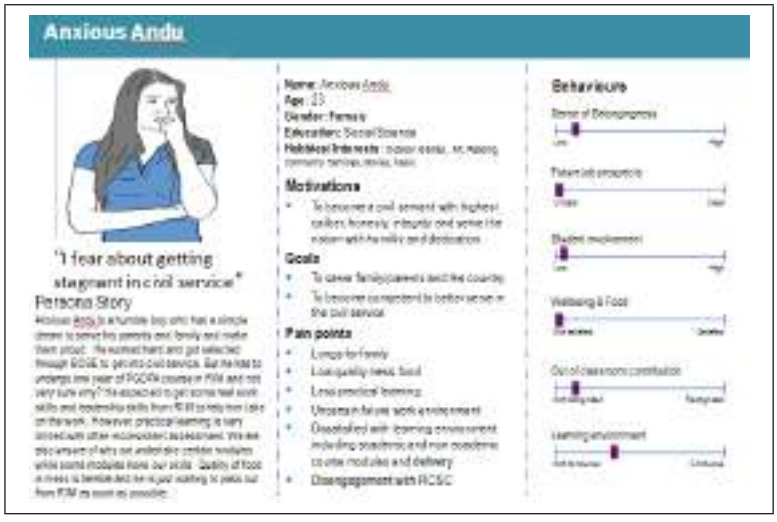
Persona Development Process



Sample Rapid Persona development Canvas.

This is used to complete step 3 of the persona development process (i.e.to describe and sketch out the Persona details).

Sample Persona



Sample Persona Presentation

TEAM BUILDING ACTIVITY

Marshmallow Challenge

Designing Thinking requires group of people to work together as a team and each team member is expected to make equal contribution. Therefore assigning specific roles and responsibilities to each team member and who should lead the team needs to be specified clearly.

Marshmallow challenge as a team building activity brings creative frame of mind and encourages innovation ideas in organization.

When to use Marshmallow Challenge

- When team members are new to each other
- To be prepare the team for the ideation phase

How to conduct Marshmallow Challenge exercise

1. Divide the team with four members each.
2. Provide the challenge materials (20 sticks of spaghetti, 1 meter of string,1 Marshmallow and 1 meter of masking tape)
3. Give the challenge "to build the tallest standing structure with marshmallow to be placed on the top of the structure within 18 minutes".
4. The following instructions are to be provided clearly:
   a. The team may use all 20 spaghetti or less and may even break it if required
   b. As much or less of the masking tape and string may be used.
   c. Eating the marshmallow shall disqualify the team.
   d. Measurement shall be taken from table surface to top of marshmallow
5. Start the challenge
6. After the measurement, the winning team to be declared
7. Conduct debrief through team reflection as to what was learnt, what role each team member played and how to improve the team performance.

PHASE III

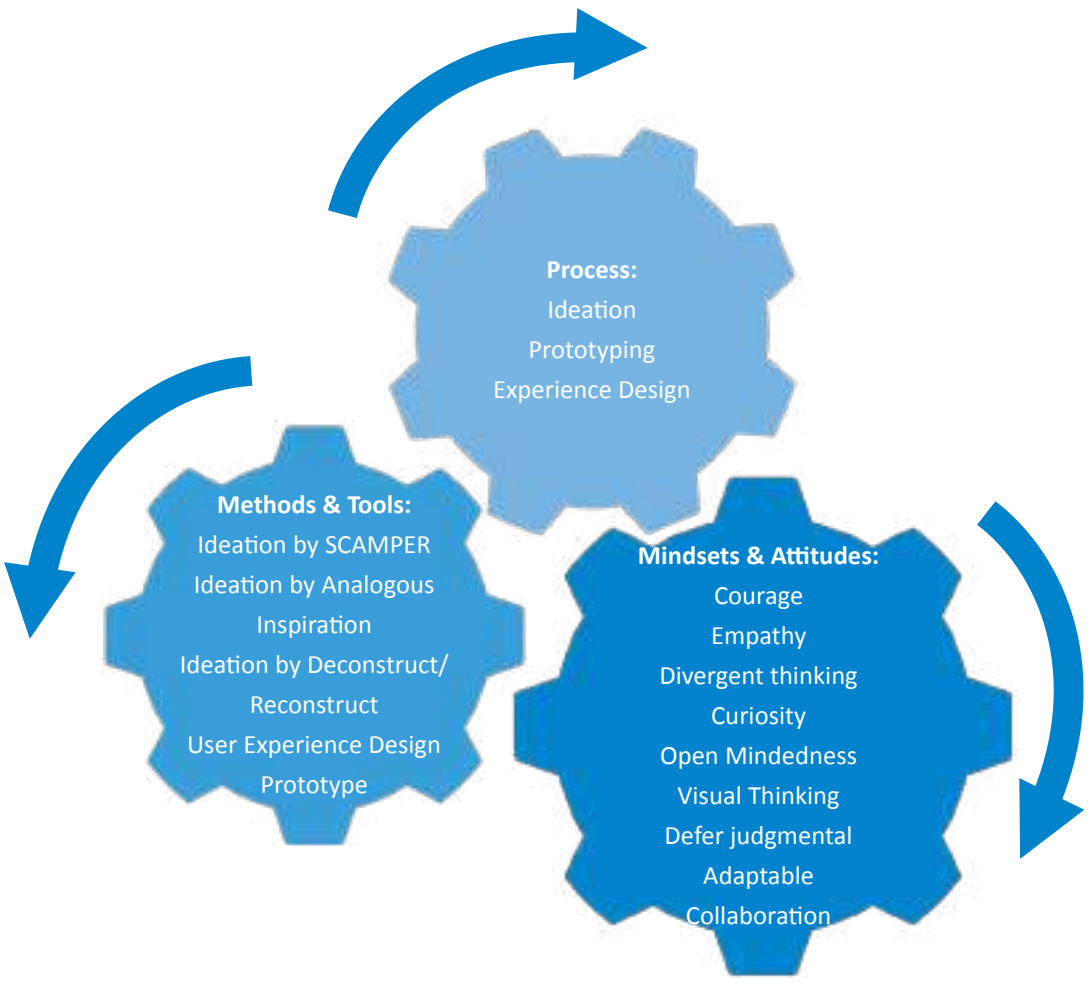
experiment

PHASE III EXPERIMENT

This is the ideation phase where huge quantity of ideas are generated by brain storming using the ideation tools. The ideas are then prototyped to provide user with ideal user experience journey.

Purpose

- To brainstorm quantity and variety of ideas around user deep needs
- To make ideas tangible and visible through building prototypes and visualizing the ideal user experience

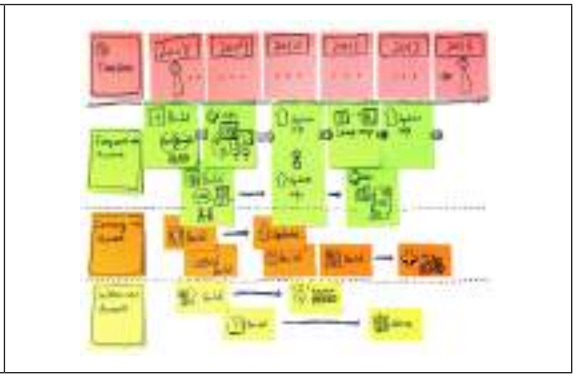


METHODS & TOOLS

Ideation by SCAMPER: A tool to generate new ideas from different perspective.

Ideation by Analogous Inspiration: Draw insights and inspiration from seemingly unrelated industry to spark creative and innovative ideas

Ideation by Deconstruct and Reconstruct



User Experience Design: It is the process of creating great experience through enhancing the user satisfaction by improving the usability, accessibility, and pleasure provided in the interaction between the user and the product/ services

Prototyping

IDEATION USING SCAMPER

SCAMPER is a creative brainstorming technique that stretches the parameters of thinking to generate new ideas from different perspective. Given any object you use SCAMPER to generate new ideas.

S
- SUBSTITUTE: What might you substitute? Who else? What else? Where else? What parts/what materials?

C
- COMBINE: What might I combine this object with? Can I combine ideas, objects, function? What this also include or do?

A
- ADAPT: What might I change? What else is this like? What does this remind me of? What might I copy?

M
- MODIFY, MAGNIFY, MINIFY: How might I change it? Bigger, smaller. What might I increase or decrease? How can I enhance or diminish attributes such as color, texture, sound, taste, smell, speed?

P
- PUT TO OTHER USE: Are there other uses of this object? What happens if I change the context or function or purpose?

E
- ELIMINIATE: What can I do without? What can I take away or remove?

R
- REVERSE, REARRANGE: What if I turned it upside down? Backwards? Inside out? What if I rearrange any parts, function or objectives?

7 Rules of Idea Generation
- Stay focused on the topic
- Go for quantity
- Be visual
- One conservation at a time
- Encourage wild ideas
- Defer judgement
- Build on ideas of others

When to use SCAMPER

When there is need to generate huge quantities and variety of ideas

How to use SCAMPER for Ideation

Step 1	• Each team to be given different objects
Step 2	• Generate the most idea using the SCAMPER canvas. There is no sequential flow while using SCAMPER.
Step 3	• As each team member has an idea, stick on the relevant column of the canvas. One idea per Post-its is used.
Step 4	• Compete to see which team generates the most ideas within the given time.
Step 5	• Within 10 minutes , 10 people are to generate 50 ideas
Step 6	• Cluster the ideas by themes

A sample SCAMPER template for ideation

SCAMPER Worksheet		
S	Substitute	(What can you substitute? What or who could you use instead?)
C	Combine	(What can you combine or bring together?)
A	Adapt	(What can you adapt for use as a solution?)
M	Modify	(Can you modify or change the item in some ways?)
P	Put to other uses	(What other or new ways could this be used?)
E	Eliminate	(What can you get rid of?)
R	Reverse	(What could you rearrange or change around?)

Ideation can also be done using POEMS

IDEATION USING ANALOGOUS INSPIRATION

What is Analogous Inspiration?

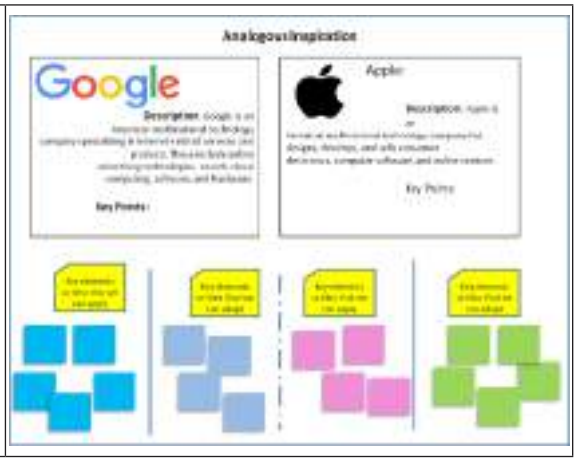
Analogous inspiration helps to draw innovative ideas from various organization and industries. It provides different perspective and prompts new and creative ideas.

When to use Analogous Inspiration

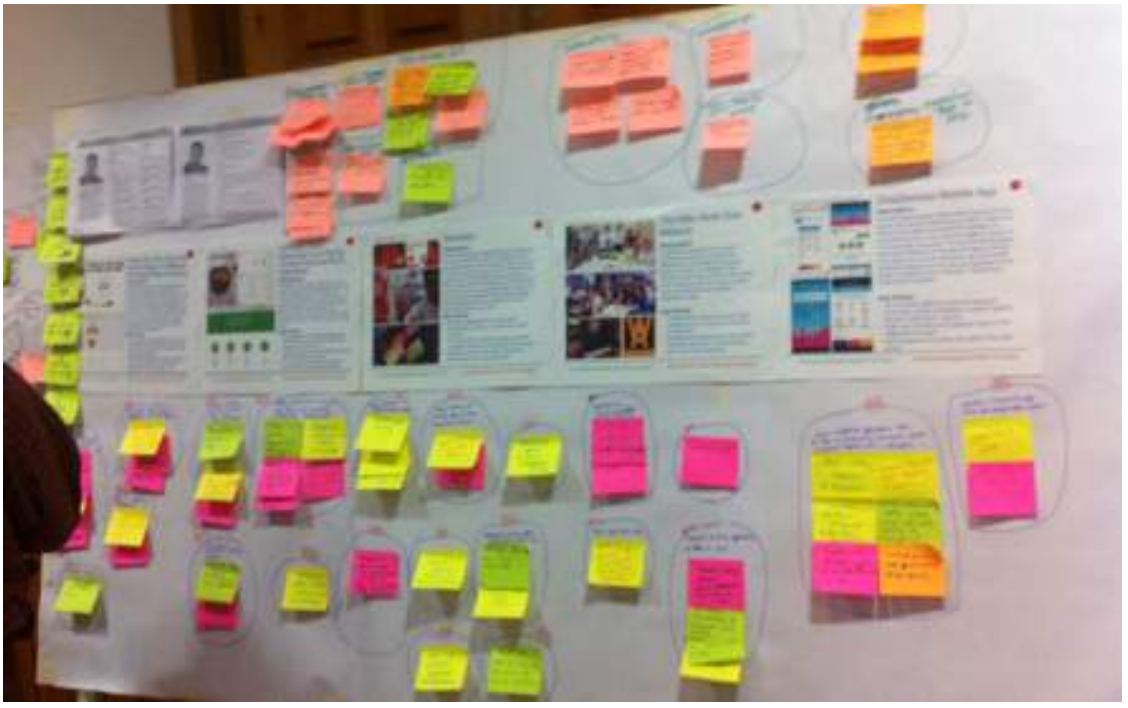
- To generate variety and quantity of ideas

How to use Analogous Inspiration for Ideation

Step 1	• Based on user deep need, each team member is to get relevant case studies by looking across other industries, organization and innovation that might serve to inspire challenge at hand.
Step 2	• Stick a large white paper on the wall and the case studies.
Step 3	• Using Post-its and pen, from the case study, every team member to identify and articulate the desirable element of good experience that you can use and adapt as per the user need.
Step 4	• As each team member has an idea, make them share and stick on the white paper.
Step 5	• Using this method, generate as many ideas as possible.
Step 6	• The ideas then are to be clustering by themes.



Example of Analogous Inspiration



Ideation using Analogous Inspiration

IDEATION USING Deconstruct & Reconstruct

Deconstruct & Reconstruct is the process of taking a product or service and initially breaking it down into its essential parts; core components; key elements, significant strengths etc and then reconstruct idea and imagine new possibilities by combining features/characteristic.

When to use Deconstruct & Reconstruct

- To generate innovative and creative ideas

How to use Deconstruct & Reconstruct for Ideation

Step 1	• Identify 5 different brands/product/Companies for five teams.
Step 2	• Using a large white paper, list minimum of 10 essential elements of the given brand/ Company by deconstructing each brand/product/company
Step 3	• The team is to discuss and select 5 essential elements and circle.
Step 4	• Using Post-its and pen, every team member is to reconstruct 2-3 imaginary new products with the 5 circled essential elements.
Step 5	• As each team member has an idea, make them share and stick on the white paper. Use one idea per Post-its.
Step 6	• Using this method, generate as many ideas as possible.
Step 7	• Draw the new product on the white paper

Example: Deconstruct & Reconstruct

Team Deconstructing & Reconstructing

Idea Generation for your Design Challenge

Having gone through all the methods of ideation/brainstorming, now it is time to generate ideas for your design challenge/ problem.

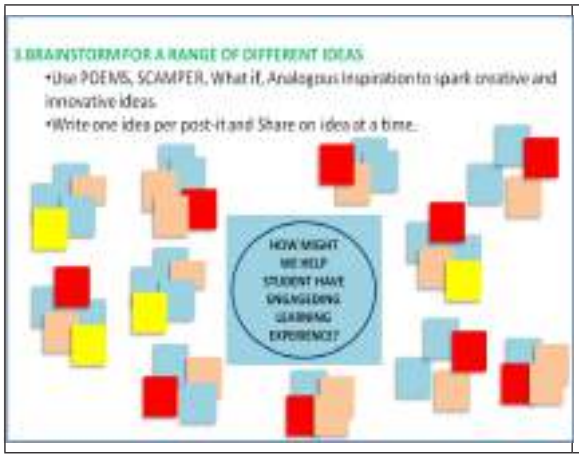
What to Do?

- Reference your persona's deep or unmet need.
- Phrase the personas' unmet need into a possibility question as: "How Might we....." Write the question on a big Post-it and place it in the middle of a large sheet of paper: "e.g. How might we make Anxious Andu feel Connected with RCSC and be Engaged in learning process while at RIM?"
- Generate quantity and variety of ideas around the person's need.
- Sketch or write 1 idea on 1 post-it. Share ideas (1 at a time), and keep generating ideas. Go for for volume. (Use the 3 + 3 + 4 method)
- Use SCAMPER, POEMS, WHAT IF or other methods to inspire and spark creative and possibility thinking.
- Pick One Idea and brainstorm in more details.
- Cluster the ideas by intention and identify themes.

All the activities pointed out in the box above are detailed in four steps as below:

Step 1

Step 2



Step 3

Step 4

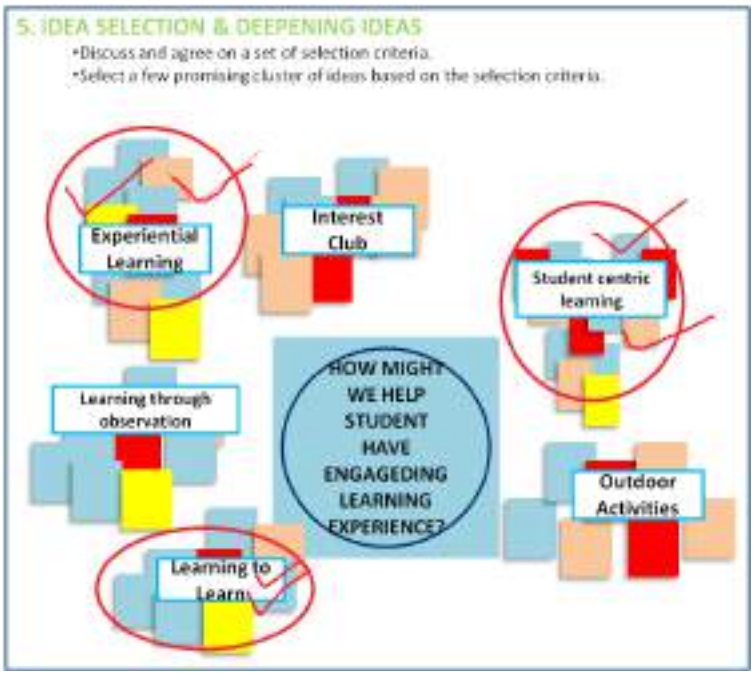
Clustering Ideas by Theme Sample

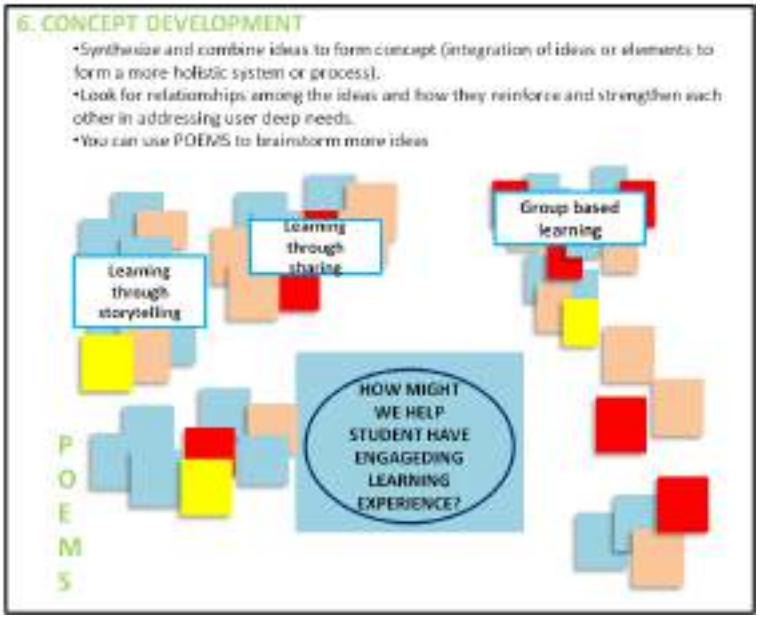
From Idea to Concept

What is Concept or Big Idea?

A concept is a meaningful and seamless combination and/or integration of related ideas to for a concept (or big idea) that best addresses the target user's deep need(s).

The concept should clearly demonstrate and articulate the target user's deep need(s) and benefits that the target user desires from you to deliver.


5. IDEA SELECTION & DEEPENING IDEAS
•Discuss and agree on a set of selection criteria.
•Select a few promising cluster of ideas based on the selection criteria.
Experiential Learning
Interest Club
Student centric learning
Learning through observation
HOW MIGHT WE HELP STUDENT HAVE ENGAGEDING LEARNING EXPERIENCE?
Outdoor Activities
Learning to Learn


6. CONCEPT DEVELOPMENT
•Synthesize and combine ideas to form concept (integration of ideas or elements to form a more holistic system or process).
•Look for relationships among the ideas and how they reinforce and strengthen each other in addressing user deep needs.
•You can use POEMS to brainstorm more ideas
Learning through storytelling
Learning through sharing
Group based learning
HOW MIGHT WE HELP STUDENT HAVE ENGAGEDING LEARNING EXPERIENCE?
P
O
E
M
S

User Experience Journey

What is User Experience Journey?

User experience design is the process of creating positive and great experiences by enhancing the user satisfaction. It is focused on the user and involves the process from how the person discovers the product/ services to how he/she uses it to achieve a goal. It outlines the details as to whether the ideas may or may not work from user perspective.

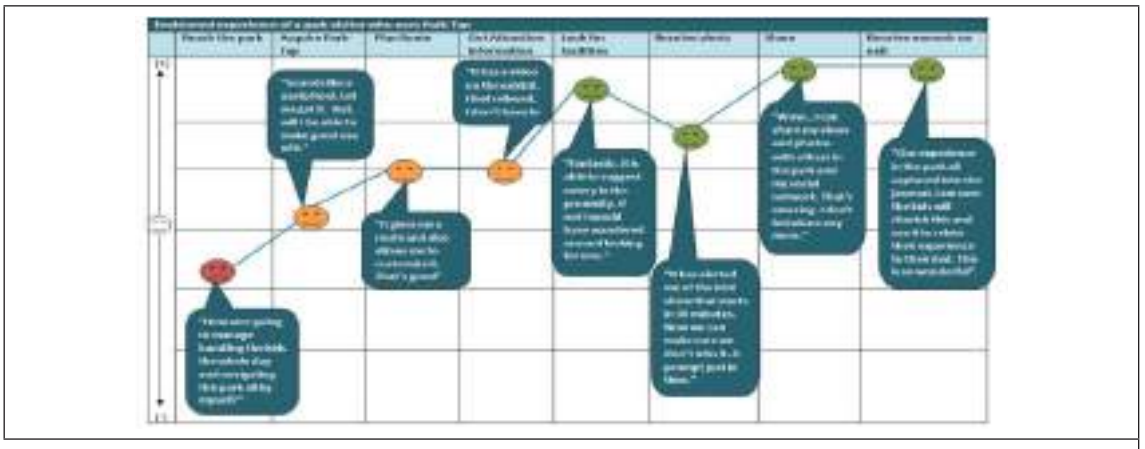
When to use User Experience Design

- To generate details of the big idea.
- To check the viability of the big ideas

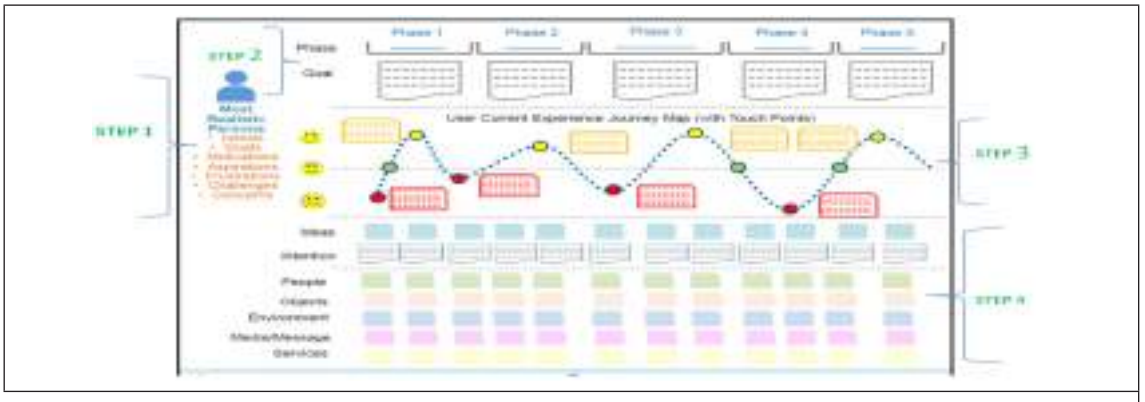
How to Map User Experience Journey

User experience journey includes the entire process from how the user discovers the product, service, process to how he/she uses it to achieve a goal. It consists of phases/stages of the behavior and touch-points across time and space. The user journey is based on user insight, deep needs, behaviors and include interactions, emotions and experiences. User Experience Journey mapping can be done in four steps:

Step 1	• Start with your most realistic persona. Review your persona and gain a deep understanding of the personas in terms of goals and objectives, aspirations and motivations, challenges and pain points, behaviours, and deep or unmet needs.
Step 2	• Identify and plot the steps or phases of the user journey. In each phases think broadly about the context, progression, objects, emotions.
Step 3	• Unfold your related ideas (from your Big Idea) into the User Experience Journey. Identify key-touch points and/or gaps in the journey where you can integrate selected ideas to address their needs, enhance their gains and/or relieve their pains. Repeat this steps a few times.
Step 4	• Deepen your ideas with POEMS framework to enrich the ideal User Experience Journey. Review each ideas being integrated into the User Journey. Envision how this idea will look like when implemented. Use POEMS framework to think through the implementation details.



Sample User Journey till step 2. Under each emotion symbol, you can give naratives to describe behaviour and emotion of the persona in this journey.



Complete User Experience Journey

PROTOTYPING

Prototyping is an approach that makes ideas and concepts more tangible and visual as compared to written ideas. It helps us to empathize with the user in terms of the viability of the solution and further minimizes the risk of failure at large scale.

How to carry out Prototyping?

Step 1	• Review the ideal user experience journey.
Step 2	• As a team, decide what are the key elements from the ideal user experience journey that you need to test with real users to get their feedback and ideas on.
Step 3	• For each of the selected key elements, determine what appropriate prototype to build. For example, if you want to test user interactions, consider putting on a skit with your team. If you are testing a logo, print it out and stick it on a t-shirt to seek feedback.
Step 4	• Keep working and improving on the prototypes with the feedback. Make it, break it, and fix it.

HEY! Lounge for youth seeking employment (MOLHR)

Prototyping BIG Ideas

PHASE IV

engage

PHASE IV ENGAGE

After your solution concept and prototypes are ready, share your ideas with the users. Give user the walkthrough experience using ideal user experience journey. Feedback and comments from the user are taken with open mind and is being reviewed to come to a common understanding about which idea works and which does not work for the user and why?Finally refinement of the solution ideas is done.

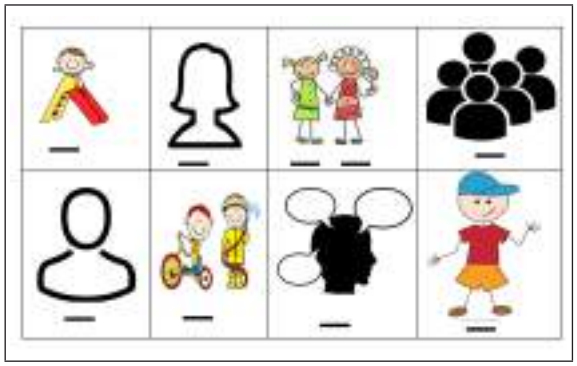
Purpose

- Ensure prototype solution matches the need of targeted users.
- Generate ideas that you may not have thought of. and weed out misfit ideas
- Identify blind-spots in our ideas.
- To fail early and cheaply.

METHODS & TOOLS

Story Telling: Giving voice to your persona's story

Co-Creation with Users: Sharing your solution ideas to the user using various methods/tools and then integrating the feedbacks and comments into the solution idea that will better meet the user's need.



Storyboarding: Visually describes series of activities of the solution idea. Each team member draws the touch-point of your idea.

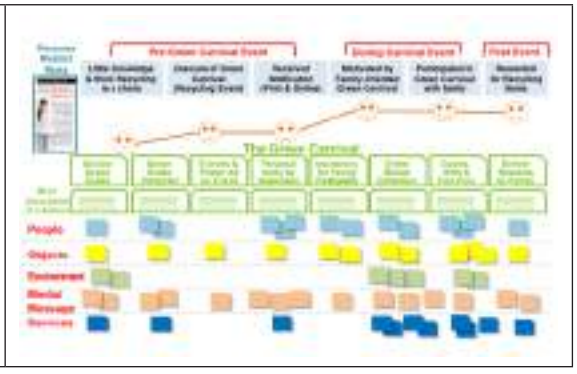
STORYTELLING

Storytelling is the most powerful way to put your ideas into the world today. Storytelling is basically giving voice to your persona's story which you have developed in the experiment phase (ideal user experience journey).

Storytelling can be done using:

- Storyboard: visual illustration of your idea
- Prototypes: developing and testing ideas at early stage before large scale implementation.
- Role Play: to act out the experience to give better understanding of your idea.

Persona



Ideal User Experience Journey

Prototypes

You have your persona, Ideal User Experience Jouryney and prototypes. Now make your solution concept more visual by storyboarding

STORYBOARDING

Storyboarding is a visual illustration of your idea for the easy and effective communication with the users. Storyboard will include your persona, need statement, solution illustrations, and impact or end result of the solution concept.

Storyboard: Visual Story

Purpose

- To graphically describe user's activities, your design idea and communicate the idea more effectively to the users.
- To create story narratives

Storyboarding your Persona's Story

BEGINNING	MIDDLE			END
The Persona	**The Setting**	**The Problems**	**The Solutions**	**The Resolutions**

Brainstorm how you intend to illustrate your persona's story in a coherent manner including the narratives into the storyboard canvas [Annexture] and then start drawing.

CO-CREATION

Co-creation involves engaging the target users early in the development process of your solution ideas in order to enhance and refine the ideas. By eliciting user's feedback and suggestions, you can better understand user needs and desires, and refine and improve solution ideas.

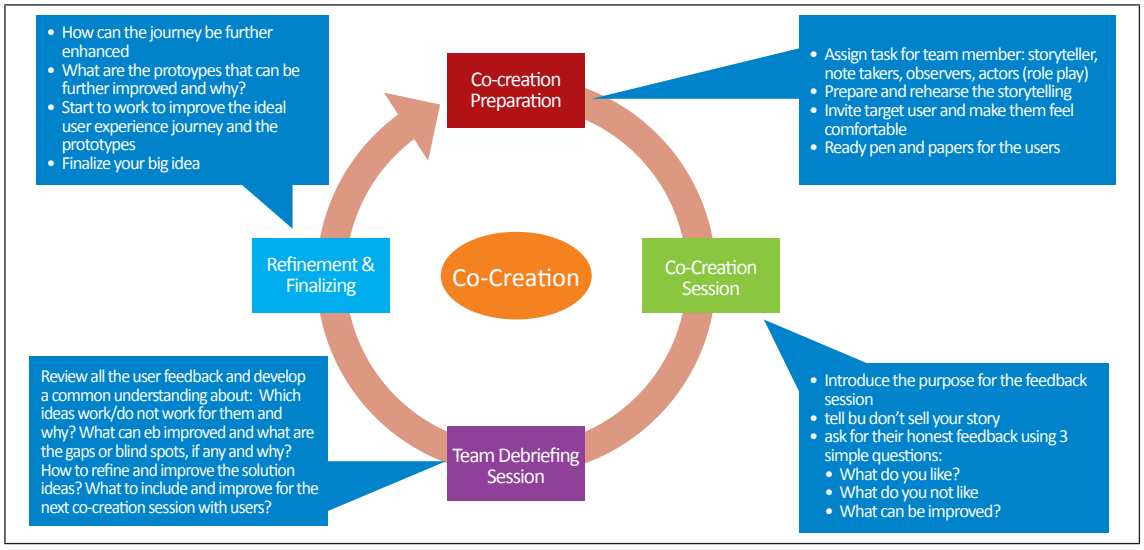
Purpose:

- Early feedback on proposed ideas and solutions;
- To better user-generation ideas and contents;
- Idea improvements;
- New creative possibilities;
- Better user acceptance and buy-in;
- Early user communication

Co-Creation with target users can be done using:

- Storyboard
- Role play
- Prototype

Steps involved in Co-creation session is show in the diagram below:



PHASE V

evolve

PHASE V: EVOLVE

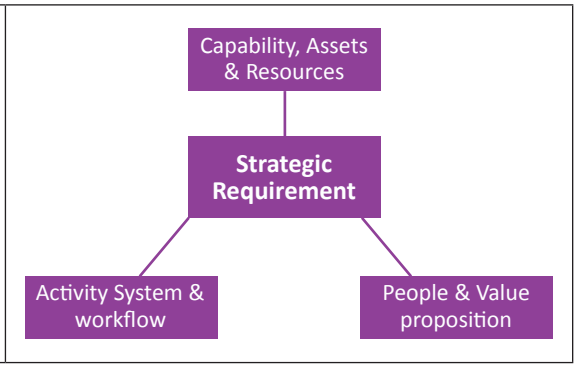
This phase involves developing smart Strategies & Implementation Plan that will deliver/achieve the Big Idea/solution deduced from earlier phases. This is the time to strategize and advance from Ideas to Reality.

Purpose

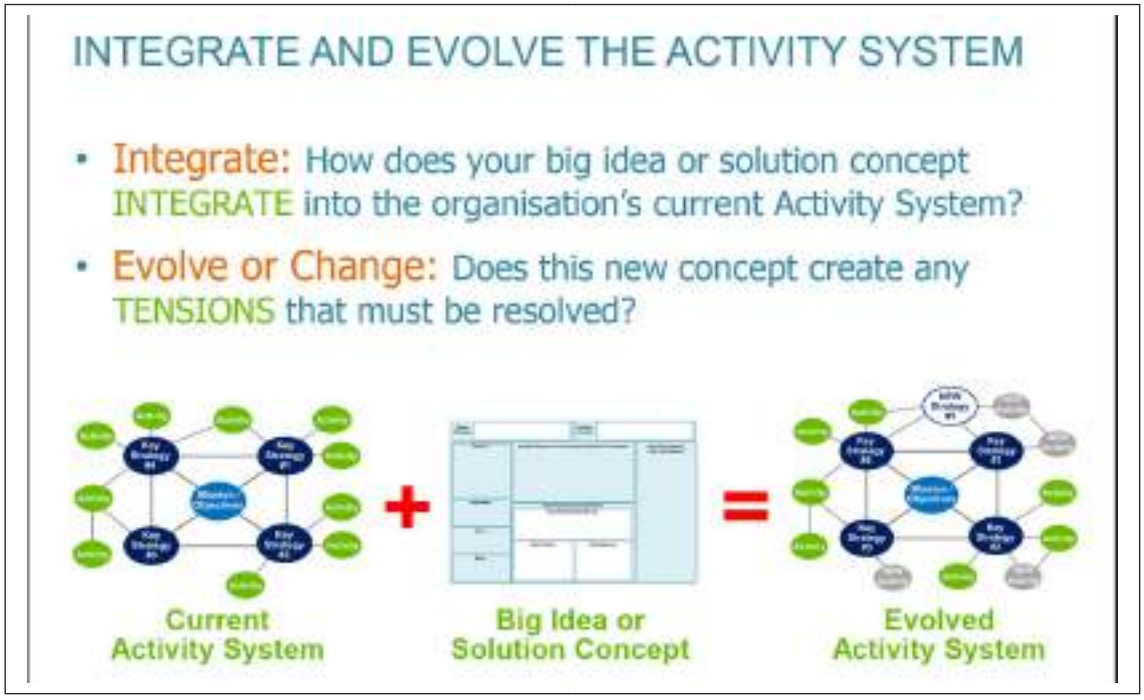
It's all about renewing your activities & strategies; here you will test some of your ideas to learn what resources are needed and how best to implement in a way that meet user needs.

- To pilot test selected user-inspired ideas to learn what works and what doesn't
- To determine the needed resources & capabilities to implement these ideas.
- To evolve the current activity system & processes to support implementation.

Concept Synthesis
Determining the Big Idea/Solution from user needs & stakeholder value propositions



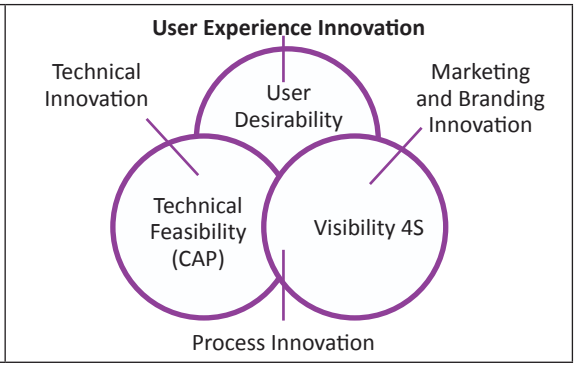
Strategic Requirements
Advancing from Ideas to Reality by assessing Organizations' Capability; Activities &People



Activity System Integration
Leveraging/Evolving/Refining the current system and process to work smarter and not harder

- Current Activity System → Solution concept → Evolved Activity System
- Evolving the process for Delivery

Viability Analysis (Impact Evaluation)
Impact assessment Using 4S



Innovation Tool
Testing solutions using user needs, CAP & 4S

Change Management (REVIEW)
Using Reasons for change; Engage staff; Visualize change; Initiate implementation; Empower people for change &Quick Winsw

Quick Wins
Think Big; Start Small & Scale Fast

Action Planning to Advance Design Challenge Project

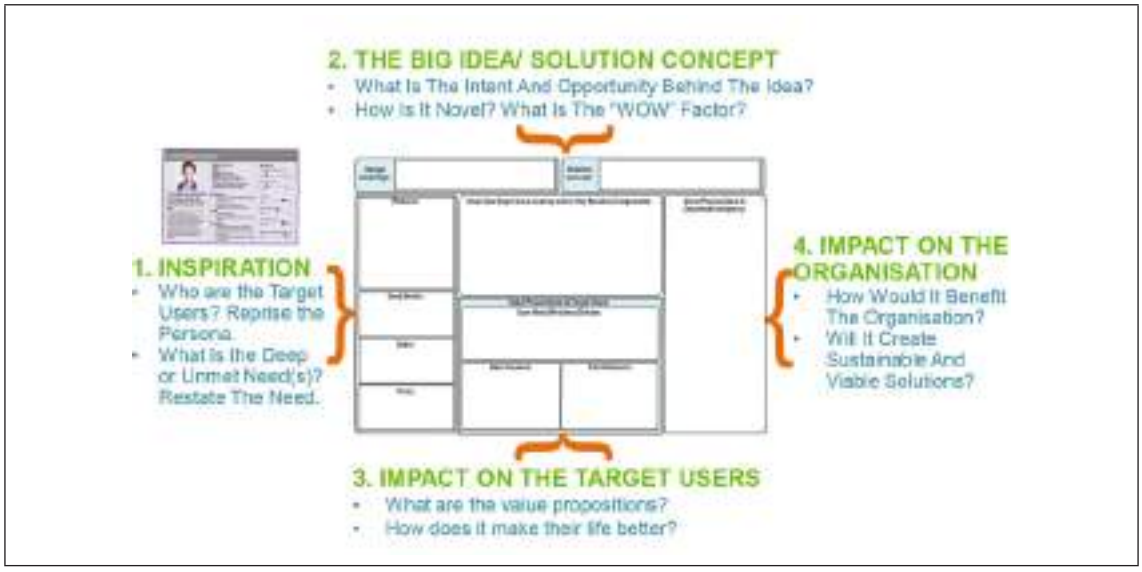
Processes

1. Recap on all the four phases
2. Synthesize the solution by reprising persona and its deep user needs; pains & gains and Value propositions
    a. Idea Refinement from the project learnings.
3. Evolving phase
    a. Integrate the New Solution by Evolving the Organization Activity System
    b. Determining strategic requirements.
    c. Identify and Design the Steps and Workflow to Delivering the Solution
    d. Evolved Activity system and workflow integration.
    e. Impact assessment of the Big Idea and Solution components.
4. Change Management (managing peoples' side of the change)
    a. Identify and implement quick wins.
    b. Action/implementation plan.
    c. Tell a compelling story of the user needs and the solution.

CONCEPT SYNTHESIS

Concept synthesis is the process of consolidating your learning (deep user needs, unmet needs and solution concept) and studying and defining the impact (value proposition in terms of function, Social, Emotional) it may have on the users and organization. Concept synthesis can be done using "11 box" tool which highlights.

Concept Synthesis is a "11 box tool" which highlights "User Centric, step by step Solution Concept" with the "Value Propositions for both parties" to the "current problem" at a glance. The following steps guide to synthesize the concept:



How to use?

Step 1	• Understand your main Users (represented by the Persona), so outline the Deep Needs; Gains and Pains in each of the boxes at the left.
Step 2	• In the top two boxes, the Challenge and the Solution Concept/idea needs to be outlined based on the opportunities it provides (the Novelty and "WOW" factor it has).
Step 3	• Next set of boxes should highlight the Key Solution Components (from where the solutions were derived) and the Ideal User Experience Journey.
Step 4	• Further, three boxes below should provide the (1) Value proposition for the User; (2) Gain Creators and (3) Pain Relievers (How it makes User's life better).
Step 5	• On the Right hand side the last box should contain the Value proposition for the Organization (Will it create sustainable and viable solutions?).

STRATEGIC REQUIREMENTS (ADVANCING FROM IDEAS TO REALITY)

Purpose

Tool which helps to analyze how feasible the Solution is based on following three critical factors that affect what an organization can and cannot do:

- Capabilities and Assets/Resources
- Activity Systems and Workflow (Processes)
- People & Values

When to use?

This should follow only after concept synthesis after determining the "Big Idea/Solution" and the different key solution components which helps to deliver or make up the Big Idea/Solution.

How to use?

Step 1	• There is a need to list down all the ideas and solutions from the concept synthesis.
Step 2	• For each key solution component what are the capabilities and resources required to deliver.
Step 3	• Then assess the existing capabilities and resources and the "GAP".
Step 4	• After the assessment of the GAP, what is the additional capability that is required to be developed and devise Strategies.
Step 5	• Assess the cost to develop the additional capability.
Step 6	• What are the external sources of expertise (potential partnership)?

Since template are available at the back of the guide book, why not we put the sample (filled ones). Not only in this phase but in all the phases.(we delete all the empty templates)

EXAMPLE: STRATEGIC REQUIREMENT

Strategic Requirements	The Big Idea or Idea Concept (Solution to be delivered) The big idea is to empower the student to discover herself; strengths, weaknesses, potentials and interest in the context of the society and culture in order to develop self-regulation skills to become a responsible and productive member of the society.		
Key Solution Components of the Big Idea	Authentic Learning Program	Student Support Program	Community Partnership
Capabilities Required to Deliver this Solution Component	• Instructional Material Development • Online I.M. Development • Authentic Performance Specialist	• Emphatic Skills • Pool of Teachers/Counselors • Registered Guidance Counselors	• Extension Workers • Communication Skills • Program/Project Proposal Developer
Current Institutional Assets & Capabilities to be Leveraged	• Printing Solutions/Equipment • IT Software/Hardware • Standardized Testing Instruments	• Guidance Centers • Training Facilities (e.g. wifi) • Student Center/Kiosks	• Faculty & Staff Extensionist • Local Government Unit (LGU) Partners
Development Strategy to Develop this Capability (if needed)	• IMD Skills Training • IT Skills Training • Authentic Assessment Tools • Product Development Skills Training	• Mentoring Training • Training of Trainers (TOT) • Infrastructure	• Community Based Programs • In-house review
Requirements and cost to Develop (High/Med/Low)	• Medium – Skills Training • High – Product Development	• Medium – Training • High – Infrastructure of Facilities	• Medium – Proposal • Medium – In-house review
External Sources of Expertise (Potential Partnerships)	• Partnership with Industries • External Linkages	• External Funds	• External Funds

ACTIVITY SYSTEM INTEGRATION

(Leveraging/Evolving/Refining the current system and process to work smarter and not harder)

Purpose

This tool shall help to implement the "Solution Concept" by integrating the solution components and the corresponding activities into the organization's current activity system and workflow. This shall help to leverage, evolve and refine the existing activity systems and processes in working smarter.

How to use?

Step 1	• Use the activity system developed in the explore phase to study the current activity.
Step 2	• Then based on the Big Idea/Solution concept, assess whether the Mission and Objectives need to be changed or not.
Step 3	• Then accordingly, assess which Key Strategies need to be changed.
Step 4	• This will then be followed by changes in processes using the tool to evolve the delivery processes.
Step 5	• In order to better deliver the solution to the users, different end to end processes (integral to the organizations) need to be streamlined. Processes can be workflow, business rules, patterns of interaction, coordination and approval procedures).
Step 6	• Specify one key solution component.
Step 7	• Then list down all the end to end processes needed to implement the above solution.
Step 8	• Assess the current process and determine gaps.
Step 9	• Incorporate in the activity systems if changes are required.
Step 10	• Assess what are the possible tensions arising from the change and needs to be resolved.

EXAMPLE: EVOLVING THE PROCESS FOR DELIVERY

BIG IDEA: **STUDENT-DRIVEN REAL-WORLD LEARNING**

Key Solution Component	Workflow / Process Needed to Implement the Solution					
	One Month Before Term Prepare Company List	One Month Before Term Prepare Project Brief	Two Weeks Before Term Print & Upload Materials online	First Lesson Team Design & Project Selection	1-Month Later Conduct Interim Student Feedback	Throughout School Term Guide & Monitor Student Learning
Component 1 Real World Student Project	Research on list of top companies which students can select. Categorise the companies by industries, market segments & prepare company briefs.	Prepare project brief such as learning objectives, selection criteria, project time-line, assessment criteria, reports and presentation requirements	Proof-read all project materials, assessment criteria, instructions, schedule, deadlines & company listing before printing & uploading online.	Conduct team-building exercise and allow students to form their own project team. Brief students on project assignment & allow them to select.	Conduct an interim student dialogue to elicit feedback, suggestions and co-create the project assignment. Address any concerns & issues raised	Provide project supervision, guidance and feedback to students on their project work and key learnings. Map against the intended learning outcomes
(Input Your Solution Component #2)						

VIABILITY ANALYSIS (IMPACT EVALUATION)

Purpose

Assessment of viability of the proposed solution is important and it is evaluated from the impact it creates on different stakeholders; potential to scale it up and how sustainable it is.

When to use?

As per logical sequencing the viability study is towards the end of all process before embarking on to the process of getting buy ins (change management) for different stakeholders.

How to use?

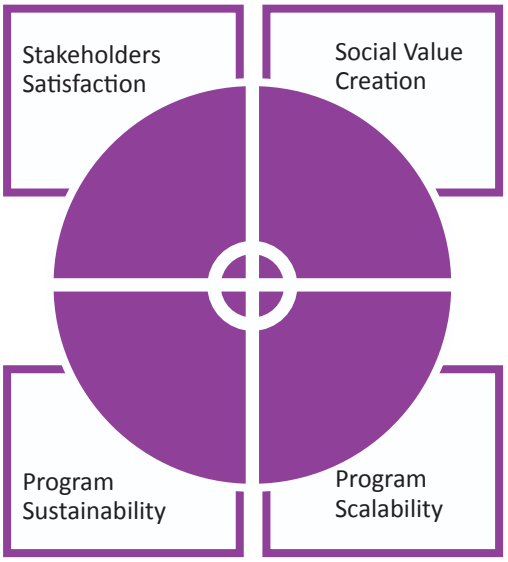
Impact assessment is carried out through the 4S tool, which should identify key indicators to indicate the following four:

Examples of Indicators:

- Awareness Level
- Happiness Rating
- User Experience
- Quality Rating
- Utilisation Rate

Examples of Indicators:

- Trust/Confidence Level
- Improvement in Quality of Life
- Effective Reach of Policy/Programme
- Closer Community Engagement



Examples of Indicators:

- Institutional Support
- Funding Stability
- Organizational Capacity
- Program Adaptation
- Joint-Operation Partnership
- Systems & Process

Examples of Indicators:

- Load scalability – increase service
- Functional Scalability – additional function
- Administrative Scalability – sharing administration
- Geographic Scalability – different location

INNOVATION TOOL USING USER NEEDS, CAP, 4S

Purpose

Innovation/transformation tool provides opportunity to understand design thinking at a glance outlining the following three key aspects:

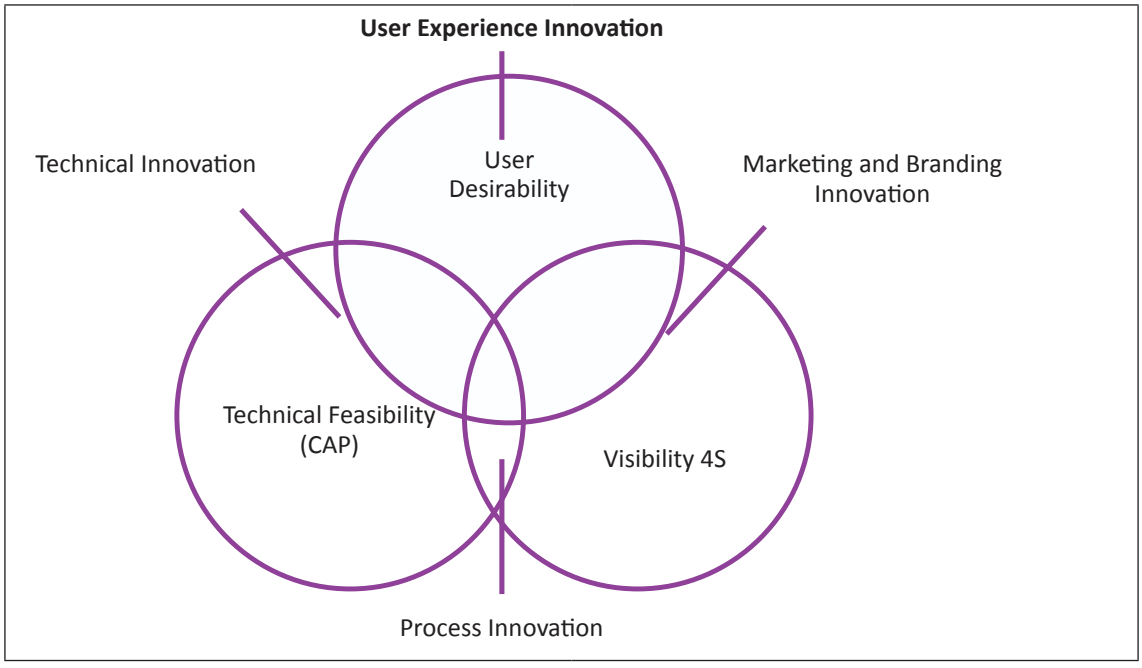
1. User desirability – basis for all unmet needs of an Organization – triggers technical, market/brand and User experience innovation.
2. Technical feasibility – assessment of current CAP and strategizing integration of solution – triggers process and technical innovation (R&D without involvement of users)
3. Viability – values created for stakeholders, social values, sustainability of the solution and scalability – triggers market/brand and process innovation

When to use?

It is useful in gaining buy-in from all stakeholders after the solution is assessed and devised towards the end.

How to use?

The tool consists of a Venn diagram, which requires outlining the three key aspects of design project and can be used to assess innovations and success/failure of the project.



Process Innovation

Capability, assets & resources
Activity systems & Workflow
People/Culture/Change/Adoption Management

Stakeholder Satisfaction
Social Value Creation
Sustainability
Scalability

CHANGE MANAGEMENT TOOL USING REVIEW

Purpose

For the new solution to be integrated in the system, some organizational shift (big or small) needs to take place. This requires Change Management skills and tools.

From many, REVIEW is one tool, which has people centric process, tools and techniques for managing people side of the change. It's a method for reducing and managing resistance to change in process, technology and organization.

How to use?

Following steps (REVIEW) are critical in leading the change:

R	Reasons for Change
E	Engage staff on the change using MARS tool
V	Visualize the vision for Change
I	Initiate the Implementation Plan
E	Empower people for the change
W	Quick Wins to motivate people to change

CHANGE MANAGEMENT PLAN FOR iACT

Design Challenge:	How Might We Help BLK Strengthen Positive Collaborative Relationship with Industry to Meet Industry's Need for Quality Employees
Solution Concept:	i-ACT (Industry Advisory Committee for Training for BLK) – provides a formalised platform that empowers industry to actively co-design & co-create quality BLK courses

What change are we making?

The roles & responsibilities of the industry advisory committee which include being empowered to co-create and co-design programmes to ensure that they are current and relevant to meeting industry's needs for quality manpower.

Why is this change important to the organisation?

- Strengthen collaboration between BLK and the industry.
- Enhance quality and relevance of BLK programmes in meeting industry's manpower needs
- Improve BLK graduates' employability

Which people and/or department will be affected by the change? How will they be affected (positive, negative, neutral)?

BLK management and secretariat will be most affected by this change. With iACT, BLK secretariat will take an active role in following up on the committee's recommendations. BLK management will need to work with iACT members to implement the recommended improvements.

How can we help the people transition (i.e. accept and make the change?)

- MOM's endorsement of BLK Secretariat new roles and responsibilities for iACT;
- Training for BLK management team on tripartite management skills
- Rewards and recognition for BLK management & secretariat for the successful implementation of iACT

What is our Change Management Plan?

Vision (Reasons) for Change	Staff Engagement	Communicate Vision for change	Implementation Plan	Empower People for the Change	Create Quick Wins
Strong collaboration & partnership between BLK & Industry leading to higher employability of BLK Graduates	Engage the BLK Management & Secretariat in identifying and selecting the industry leaders for iACT.	MOM to identify and inform the BLK management early of this iACT initiative and invite BLK to participate.	Pilot test the iACT initiative in 1 BLK	Empower the BLK Management & Secretariat to plan the iACT agenda & implement the recommended improvements.	To set up iACT in one BLK where the recommended industry leaders are appointed as its committee members

QUICK WINS

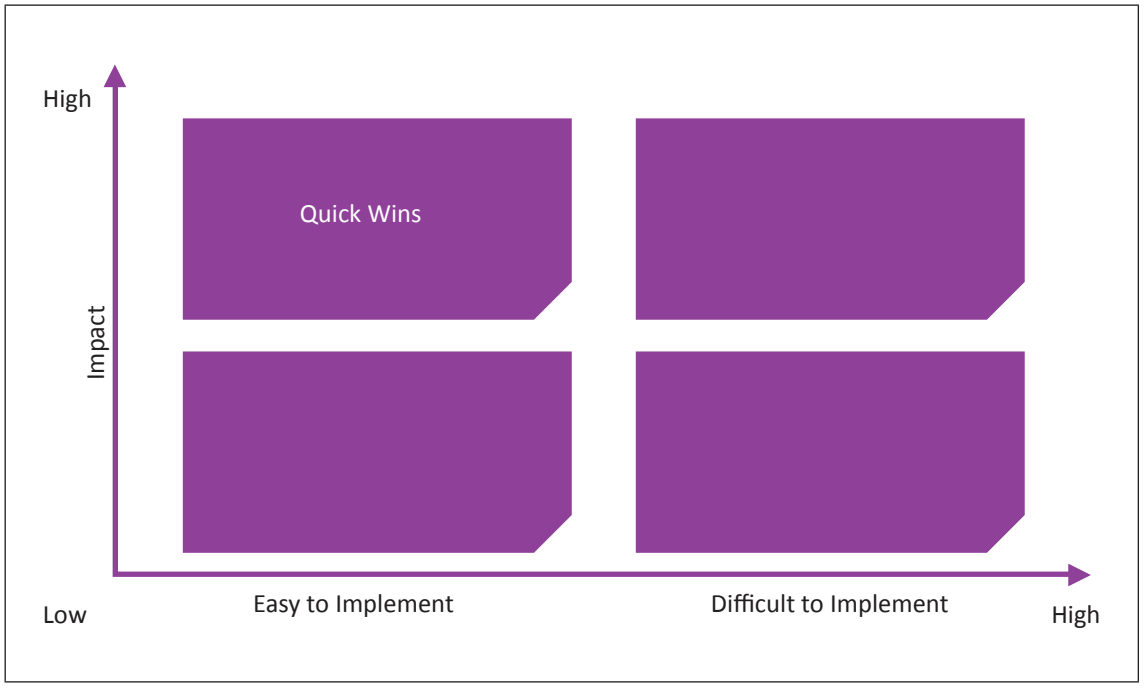
Quick wins are early small successes which are relatively cheap and easy; improvement is visible; with immediate benefit and can be implemented quickly. It helps secure organizational support and builds momentum to advance the Big Idea forward.

When to use?

First few baby steps in jump starting the implementation of the Big Idea.

Identifying and Selecting Quick Wins

- Review the solution components and key activities that lead to the Big Solution.
- Criteria for selecting impact should be based on Value creation; stakeholder satisfaction; sustainability and scalability.
- Plot this on the matrix below depending on the effort and resources required to implement and what kind of impact it has.
- Select at least 2 – 3 Quick Wins and identify success indicators; who will lead and the time frame.



Planning and Implementing Quick Wins

Once Quick Wins have been identified and agreed to, plan, schedule and acquire the necessary resources to develop and implement them. Planning and implementing Quick Wins include:

Step 1	• Develop, document and get agreement on Quick Wins
Step 2	• Clearly define success indicators of Quick Win and how it can be measured
Step 3	• Identify and assign people who will lead the implementation
Step 4	• Define time-line and key milestones of Quick Win implementation
Step 5	• Set status update timetable
Step 6	• Communicate results of Quick Win

QUICK WIN #1: (Setting up of iACT)

What is this Quick Win (1) about?	- Programme & recomentation of i-ACT - Set up the Secretariat for i- ACT - List of recommended Industry Leaders for iACT industry committee members
What are the success indicator(s)? How would it (these) be measured?	The recommended industry leaders are appointed as i- ACT committee members for BLK training.
What are the resources and/or staff trainings needed?	- Staff to operate the secretariat - Operating budget - Office and equipment
Who will lead this Quick Win implementation?	Head of BLK Lembang 1. Mr. Nono 2. Mr. Edi
What are the key steps needed to implement this Quick Win? What is the time-line till completion?	Set up i- ACT secretariat appointment letters from MOM
When will be the status or progress update?	(1-2 week) identity staff office budget for i- ACT, (3 WEEK) approvad i- ACT stat office & budget, (4 week) lounihing i- ACT secretriat, (5 -6 week) identity for i- ACT member ,(6-7 week) selection of i- ACT member & approved from mom, (7-8 week) meeting of i- ACT member working ...one years,(8-9 week) meeting on teaching & learning material of BLK , 9-10 WEEK) drafting recomendation of mom, (11-12 week) reporting to mom & member of to i- ACT
When will this be completed?	3 month
How would the Success be communicated?	DIRECTOR OF GENERAL TRAINING & PRODUCTIVITY, SECRETARIAT DIRECTORATE GENERAL OF TRAINING & PRODUCTIVITY (MR. Kunjung)

ACTION PLANNING TO ADVANCE DESIGN CHALLENGE PROJECT

Purpose

This is the action planning stage to distill, adapt and integrate to revise the implementation action plan for selected solutions and quick wins.

When to use?

This is the second last lap of the entire design project before seeking buying in from the stakeholders.

How to use?

Step 1	• As a team discuss and plan how your team intends to bring your project forward into pilot testing and implementing some of your ideas.
Step 2	• Your plan should clearly detail the what-why-who-how-and when in taking your project forward using the below template.

ART OF STORY TELLING

This is the final stage of the design project to tell a compelling story of transformation to persuade audience to Think Different with assessed user needs; solutions; plans for integrating solutions in the activity system and the way forward.

How to plan to tell a compelling story?

a. Begin with the Wall

b. Know your Audience

c. Set up your hook Early

d. Have a compelling plot

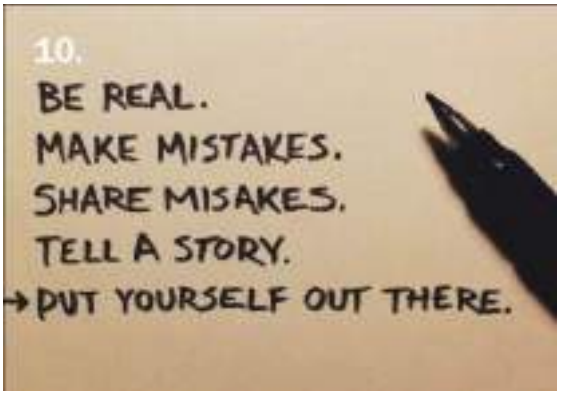
e. Have a clear theme

f. Remove the Clutter

g. Highlight the problem

h. Demonstrate a clear change

i. Make them feel



j. Be Real

Practise, Practise, And More Practise

TEMPLATES

EXPLORE PHASE

STEEP ANALYSIS

SOCIAL & DEMOGRAPHICS

- .....
- .....
- .....
- .....
- .....

TECHNOLOGY

- .....
- .....
- .....
- .....
- .....

ECONOMY

- .....
- .....
- .....
- .....
- .....

ENVIRONMENT & NATURE

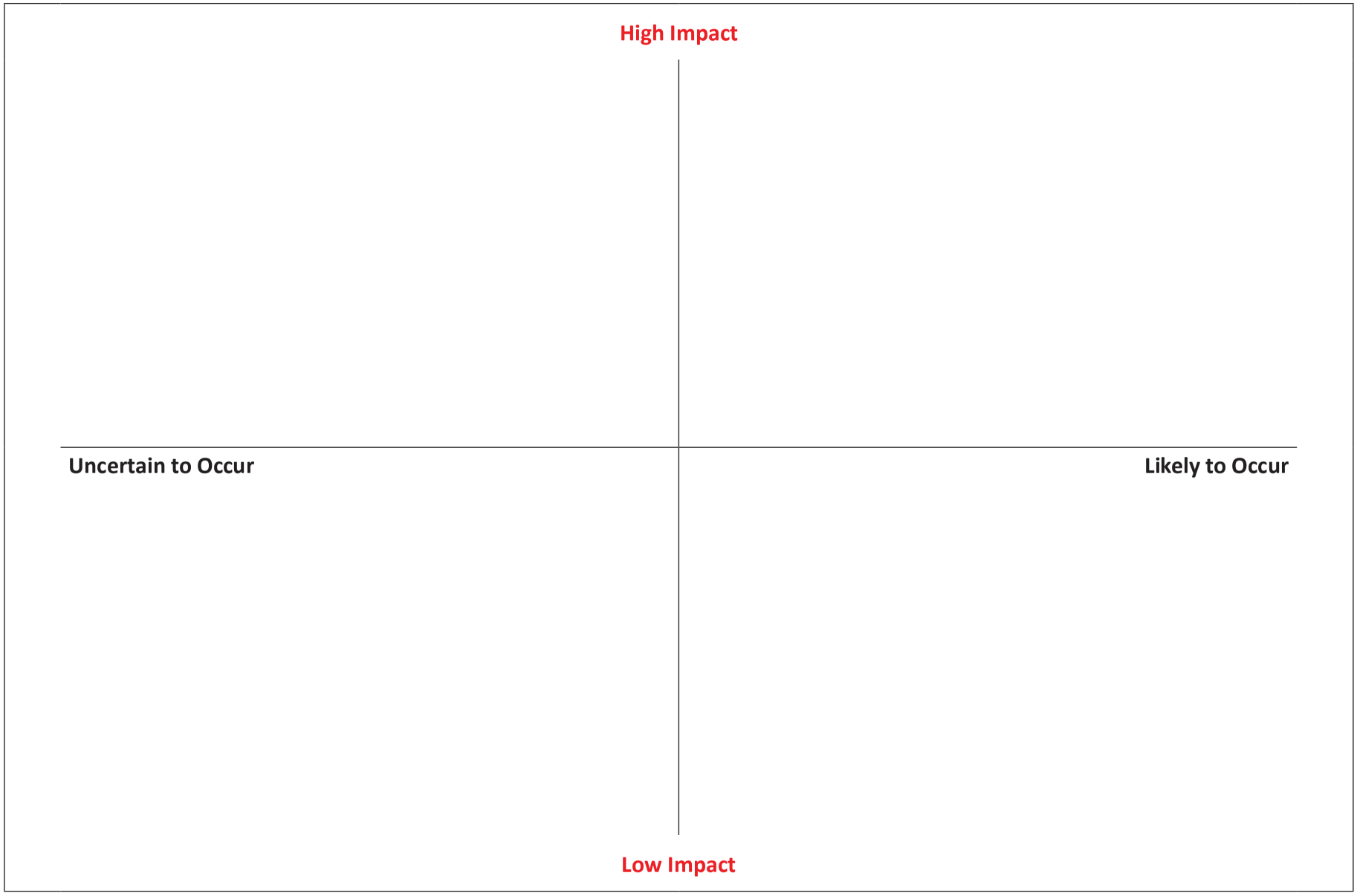
- .....
- .....
- .....
- .....
- .....

YOUR DESIGN CHALLENGE ISSUE

POLITICS & LEGAL

- .....
- .....
- .....
- .....
- .....

STEEP ANALYSIS MATRIX



STEEP ANALYSIS PRIORITIZATION

Discuss 2 key trends from the High Impact - Likely to Occur quadrant:

1.

2.

Discuss 2 treds from the High Impact - Uncertain to Occur quadrant

1.

2.

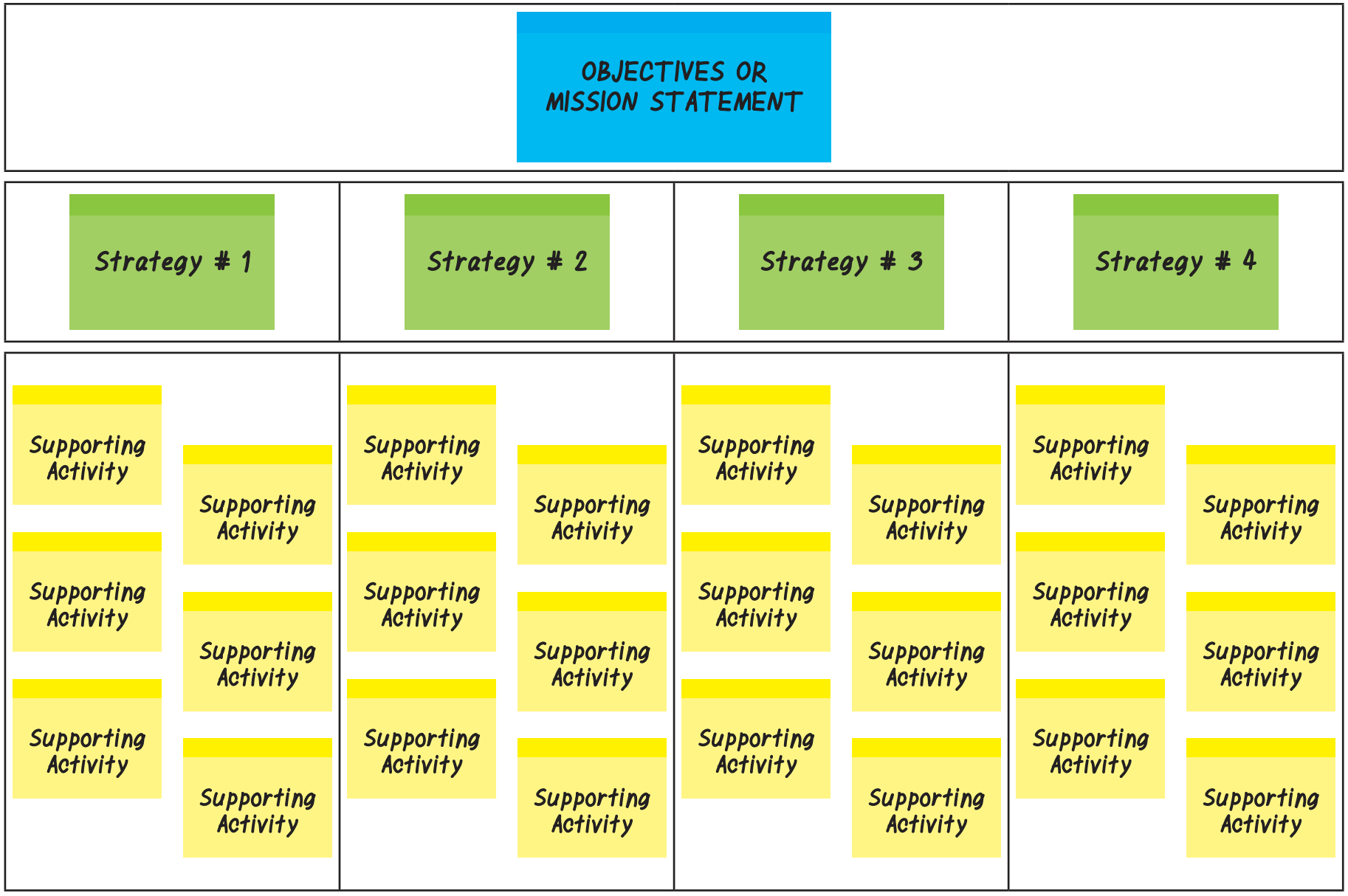
STEEP ANALYSIS MATRIX

	URGENT	LESS URGENT
IMPORTANT		
LESS IMPORTANT		

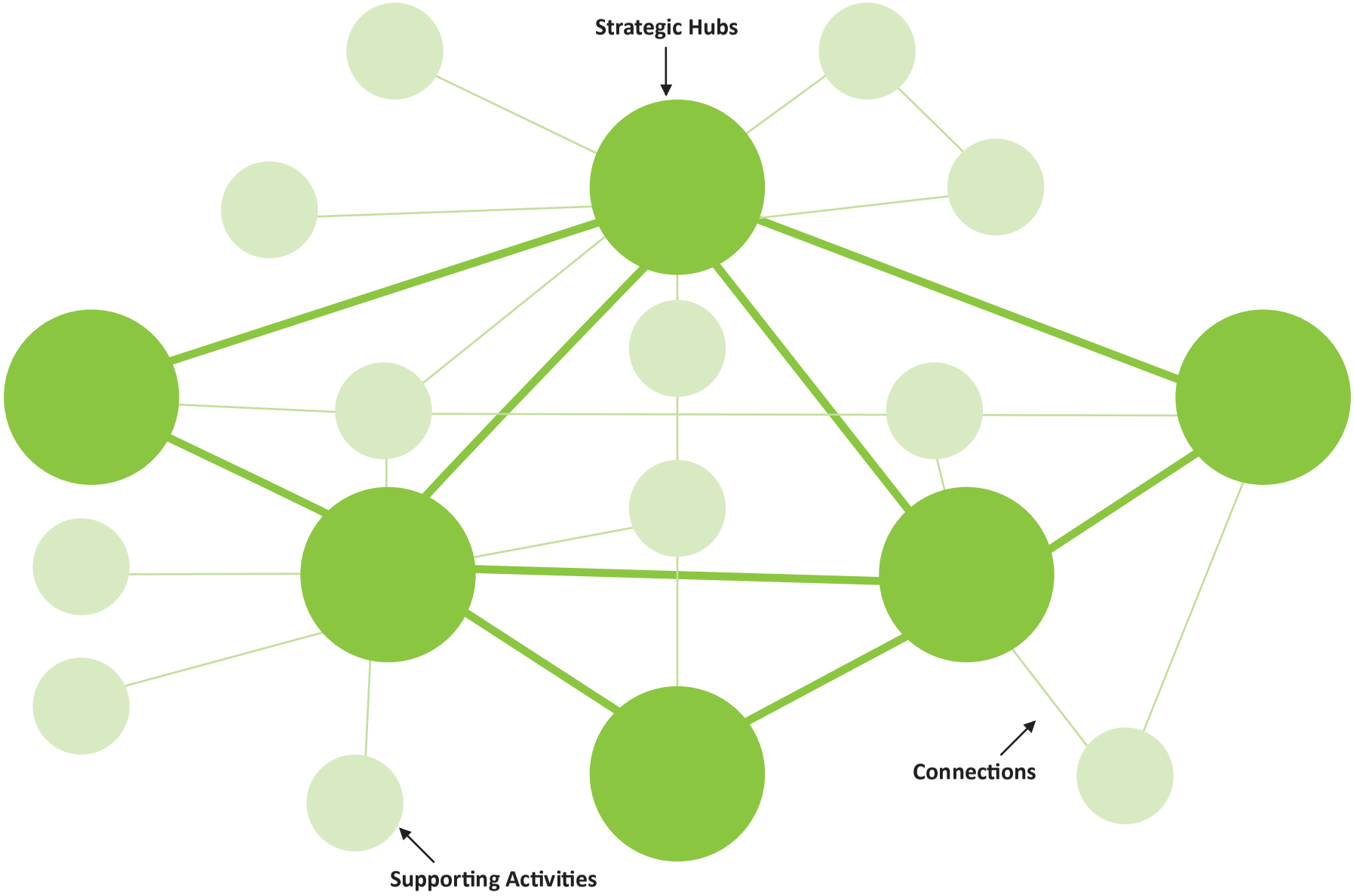
SYNTHESIS: MAKING SENSE OF STEEP ANALYSIS AND STRATEGIC PRIORITIES

Assessment Questions	Synthesis: Sense Making
What relationships among the trends do you perceive? How are they related? Why are these relationships important	
What opportunities and/or challenges need immediate attention going forward for your design challenge? And why?	
What would it take to create positive change on this issue relating to your design challenge?	
Who else would be interested in this issue? Why should they care? What conversations would you have with them?	

MAPPING ORGANIZATION ACTIVITY SYSTEM

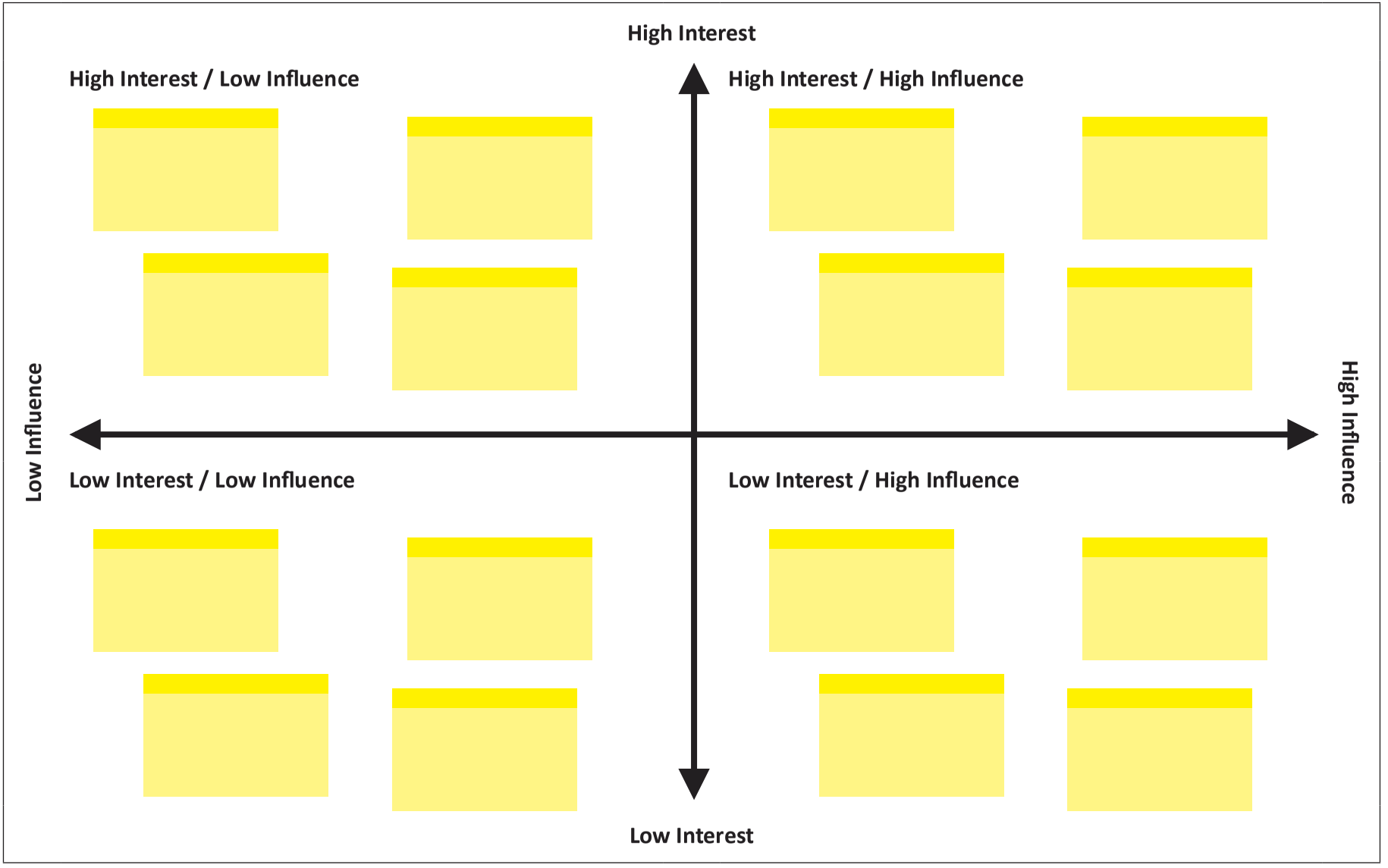


KEY COMPONENTS OF ACITIVITY SYSTEM



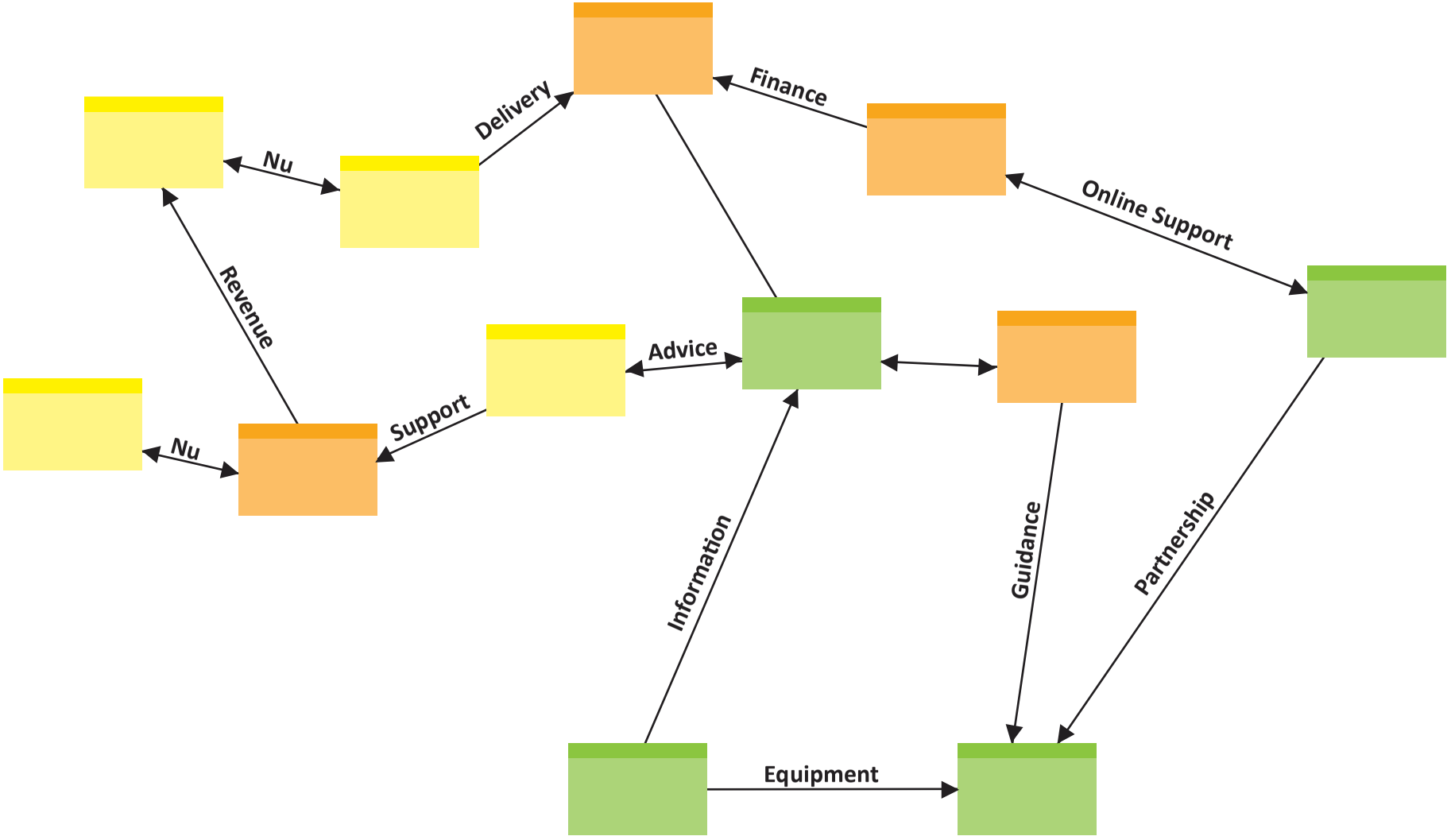
STAKEHOLDER MAPPING MATRIX

(Identify your relevant key stakeholders)



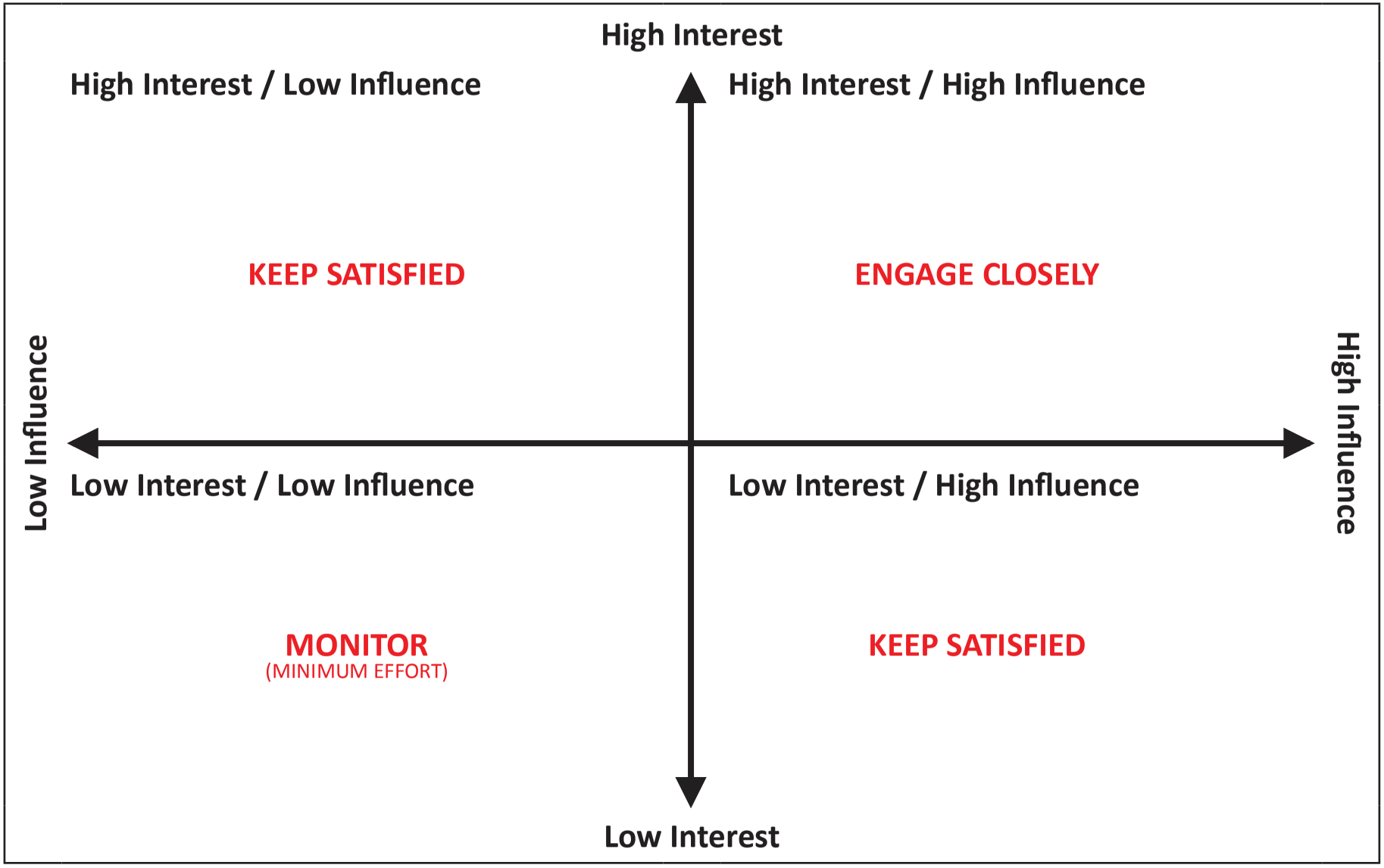
STAKEHOLDER LINKS & RELATIONSHIP MAPPING TEMPLATE

(Define the links and relationships between stakeholders)



STAKEHOLDER PRIORITY MAPPING MATRIX

(Develop engagement Strategies)



STAKEHOLDER ANALYSIS & ENGAGEMENT STRATEGY

Key Stakeholders	Relationships	Skateholder's Interest(s) in the Design Challenge	Impact Assessment	Strategies to Gain Support or Reduce Obstacles

PROJECT BRIEF AND OPPORTUNITY FRAMING TEMPLATE

A. CHALLENGE OR PROBLEM DEFINITION		
Project Sponsor	Organization Name	
	Address and Contact	
	Contact Person(s)	
Project Title		
Design Challenge		
Design Challenge Context and Background Info	What are the issues and opportunities that inspired this Design Challenge	
	Why does this Design Challenge matter to the organization	

PROJECT BRIEF AND OPPORTUNITY FRAMING TEMPLATE

A. CHALLENGE OR PROBLEM DEFINITION (contd...)	
Goals and objectives of the Design Challenge	
Target users of this Design Challenge	
Other key stakeholders	
Previous efforts to solve this Design Challenge	
Current activity system and process	

PROJECT BRIEF AND REFRAMING PROJECT CHALLENGES TEMPLATE

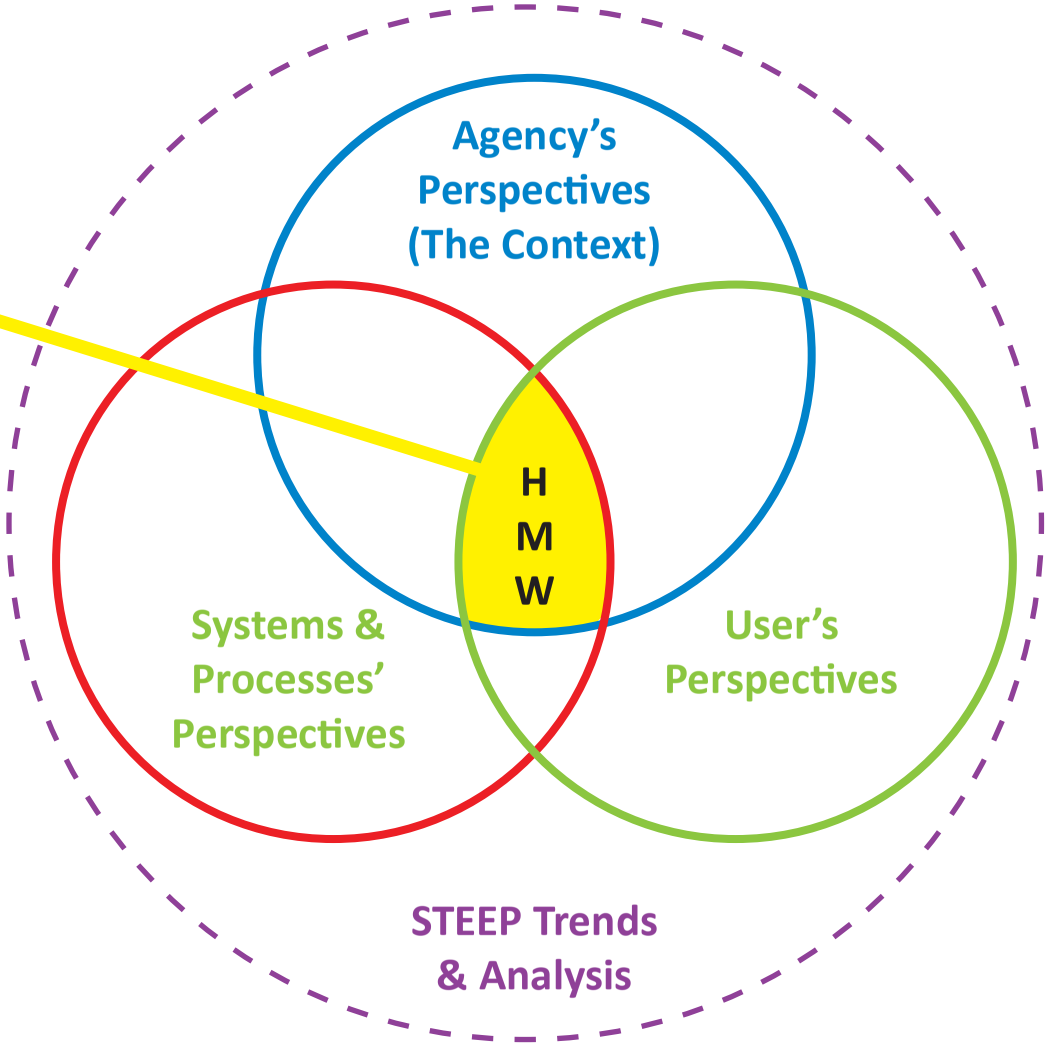
B. OPPORTUNITY FRAMING	
Real issues behing this Design Challenge	
Inspirations from others in solving this Design Challenge	
Teams contributions	
Success criteria	
“HOW MIGHT WE” Opportunity/ possibility statement	

REFRAMING THE OPPORTUNITIES TEMPLATE

Reframe your opportunities and filter one

"How might we.....?"

1. "How might we.....?"
2. "How might we.....?"
3. "How might we.....?"



EMPATHISE PHASE

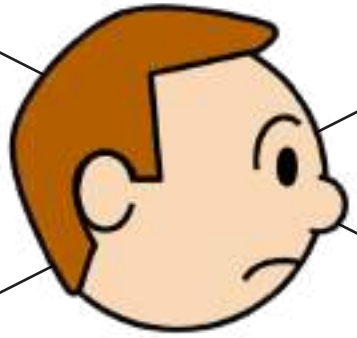
POEMS FRAMEWORK TEMPLATE

Location:	**Date/Time/Period :**	
Report By:		
POEMS Framework	**Field Visit and Onsite Observation**	**Insights**
People: Who are involved? (e.g. commuters, street vendors, office workers, children, motorists, delivery men, etc.) What roles do they play? How are the people engaging with each other? How are they related? What is the relationships? What is the social context?		
Objects: What artifacts are important? (e.g. bus stops, signage, traffic lights, benches, etc.) What roles do they play? How are people engaging with the objects, and with their surrounding? What is interaction? How does the object influence people's behavior?		
Environment: Where is the action taking place? (e.g. public bus, road side, government office, garden, etc.) What is happening? What are the people doing? How do the people behave in this environment? How does the environment influence people's behavior? What is the mood? Ambience?		
Messages & Media: What are the messages and communication media used? (e.g. signage, online materials, posters, apps, etc.) What roles do they play?		
Services: What are the services and support systems provided? (e.g. registration, library services, hands-on guide, online booking, etc.)		

GENERATE INTERVIEW QUESTIONS

- What does the user think & feel?
- What is really important to the user?
- What drives the user?
- What worries the user?
- What are user's aspirations

- What does user hear from others?
- What do friends and family say?
- Who influence the user and how?
- What communication channels are influential>



- What does user see?
- What does it look like?
- Who surrounds the user?
- Who are the user/s friends?
- What makes the user happy? Sad? Proud?
- What problem does the user encounter?

- What does the user say? Do?
- What is the user attitude?
- What could the user be telling others?
- Are there conflicts between what the user say/do and truly think & feel?

PAIN?	• What barrier and challenges does the user face? • What are the user's frustrations? • What obstacles are there? • What makes user feel indifferent?	**GAIN?**	• What makes user happy or satisfied? • What does the user truly want? • How does the user measure value? • What gives user passion/motivation?

EMPATHY MAP & USER JOURNEY TO GENERATE INTERVIEW QUESTIONS

DOING	SEEING	HEARING/SAYING	FEELING/THINKING	FRUSTATION	NEEDS/WANTS
When does target User go there? How does target User go there?					

USER INTERVIEW NOTES

Interviewer Name	
Note Taker & Observer	
Interviewee Name	
Interviewee contact details	
Date/Time of Interview	
Interview Location	
Interview's background information	
Gender	
Age Range	
Profession	
Education Background	
Family	
Hobbies	
Likes / Dislikes	
Interview Notes (Verbatim)	**Observations/Quotes**
	Page 1

POST INTERVIEW DISCUSSION: ABOUT THE INTERVIEWEE

Interviewee's Goals &/or Motivation	Interviewee's Aspirations:
Interviewee's Current Experience	Interviewee's Challenges & /or Pain Points
3 most memorable things about the interviewee:	User Insights &/or Needs

POST INTERVIEW DISCUSSIONS: ABOUT THE PROCSS

What went well with the Interview?
What did not go well with the Interview?
How can we do better for next Interview?

END OF INTERVIEW NOTES

POST INTEREVIEW DEFREIF PRESENTATION:

Summary profile of interviewee

Insert Photo Here	**Overview of interviewee** (brief demographic profile, likes/dislikes, lifestyle etc..,):

Summary profile of interviewee

Goals/Motivation:	**Aspiration:**
Current Experience:	**Challenges and Pain Points:**
3 most memorable things about the interviewee:	**User Insight & Deep Needs:**

DOCUMENTING INSIGHTS & NEEDS: EXAMPLE

INSIGHT # :

Insert narrative of the user insight, needs, quotes here.

Need:

Quotes1:

Quotes2:

PERSONA CANVAS	Persona Name:	
Demographic Profile: Age: Gender: Home: Family: Education Background:	Goals:	Deep Need Statement:
	Motivation/Aspiration:	Diversity of Needs:
Hobbies/Likes/Dislikes:	Challenges/Pain Points:	
Social & Family Lifestyle:	Behavior:	

EXPERIMENT PHASE

SCAMPER WORKSHEET

S	**Substitute**	(*What can you substitute? What or who could you use instead?*)
C	**Combine**	(*What can you combine or bring together?*)
A	**Adapt**	(*What can you adapt for use as a solution?*)
M	**Modify**	(*Can you modify or change the item in some ways?*)
P	**Put to other uses**	(*What other or new ways could this be used?)*
E	**Eliminate**	(*What can you get rid of?*)
R	**Reverse**	(*What could you rearrange or change around?*)

RECONNECTING WITH OUR PERSONAS

Behaviors	Aspirations	Motivations	Challenges	Pain Points

User Insights	Deep User Needs	Constrains &/or Design Criteria

ENGAGE PHASE

STORY BOARD CANVAS

Directions:

- Review your persona, the ideal user experience journey and the prototype you have created.
- Brainstorm how you intend to illustrate your persona's story in a coherent manner including the narratives.
- Use the storyboard template/canvas provided and start drawing.

Title: _____

STORY BOARDING CANVAS

BEGINNING	MIDDLE			END
The Persona	The Setting	The Problems	The Solutions	The Resolutions

EVOLVE PHASE

STRATEGIC REQUIREMENT TEMPLATE

Strategic Requirements	The Big Idea or the Idea Concept (Main Solution to be delivered)		
Key Solution Components of the Big Idea			
Capabilities Required to Deliver this Solution Component			
Current Organizational Assets & capabilities to be Leveraged			
Development Strategyto develop this capability (if needed)			
Requirements and cost to develop (high/med/low)			
External Sources of Expertise (potential partnership)			

EVOLVING THE PROCESS FOR DELIVERY

Key Solution Component	Workflow/Process Needed to Implement the Solution					
(Input Your Solution Component 1)	Process 1 (Header)	Process 2 (Header)	Process 3 (Header)	Process 4 (Header)	Process 5 (Header)	Process 6 (Header)
	Process 1 (Description)	Process 2 (Description)	Process 3 (Description)	Process 4 (Description)	Process 5 (Description)	Process 6 (Description)
(Input Your Solution Component 2)						
(Input Your Solution Component 3)						
(Input Your Solution Component 4)						

IMPACT EVALUATION INDICATORS

Criteria	Indicators & Measurement	Stakeholders
Social Value Creation	1. ... 2. ...	1. ... 2. ...
Stakeholder Satisfaction	1. ... 2. ...	1. ... 2. ...
Solution Sustainability	1. ... 2. ...	1. ... 2. ...
Solution Scalability	1. ... 2. ...	1. ... 2. ...

ACTION PLANNING TO ADVANCE THE DESIGN CHALLENGE PROJECT

Idea What idea for implementation	**Objectives** Why is this idea important? Values and benefits	**Responsibility** Who will lead this?	**Implementation** How will this be implemented	**Resources** What capability and resources are needed?	**Completion** When will this be completed?

IDENTIFYING QUICK WIN

What is this Quick Win (1) about?	
What are the success indicator(s)? How would it (these) be measured?	
What are the resources/staff trainings needed?	
Who will lead this Quick Win implementation?	
What are the key steps needed to implement this Quick Win? What is the timeline till completion?	
When will be the status or progress update?	
When will this be completed?	
How would the Success be communicated?	

CONCEPT SYNTHESIS

<table>
<tr><td>Design Challenge</td><td colspan="2">How Might We...</td><td>Solution Concept</td><td colspan="2"></td></tr>
<tr><td colspan="2">Persona:</td><td colspan="3"></td><td rowspan="4">Value Propositions to Organizations/Agency:</td></tr>
<tr><td colspan="2">Deep Needs:</td><td colspan="3">Value Propositions to Target Users:</td></tr>
<tr><td colspan="2">Gains:</td><td colspan="3">User Need (Problem) Solvers:</td></tr>
<tr><td colspan="2">Pains:</td><td colspan="2">Gain Creators:</td><td>Pain Relievers:</td></tr>
</table>

M – A – R – S FRAMEWORK

Use the MARS framework to understand the people's behavior in the face of the change and innovation **Motivation** It is about the **Why**/the **Will** to change					
		Ability	**Role**	**Systems**	
		It is about the **Tools** and **Skills**	It is about defining **Roles/ToRs**	It is about the **Support** system **(Resources)**	
Engage team in conversation to connect and to empathise. Listen to welcome the truth and to gain trust	**Think**				
	Feel				
	Do				

WHAT IS OUR CHANGE MANAGEMENT PLAN?

Vision (Reasons) for change	Staff Engagement	Communicate vision for change	Implementation Plan	Empower people for change	Create Quick Wins

Master Trainers

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Dy. Executive Engineer
Ministry of Works and Human Settlements

SONAM JAMTSHO
Architecture
Thimphu Thromde

THINLEY GYELTSHEN
Asst. Tourism Officer
Tourism Council of Bhutan

JASRAJ LIMBU
Assistant Labour Officer
Thimphu Thromde

PHURBA TSHERING
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